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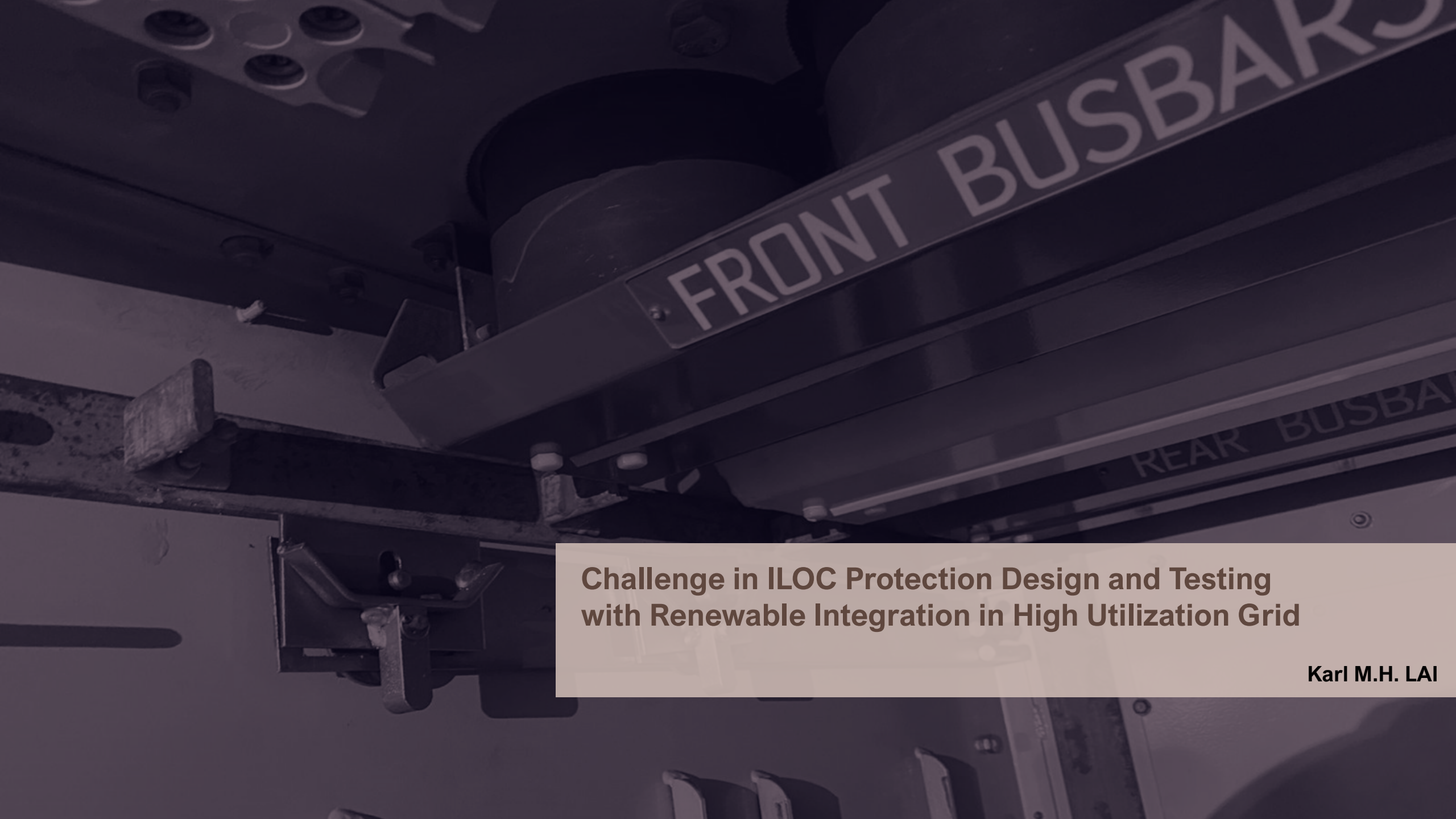
Challenge in ILOC Protection Design and Testing with Renewable Integration in High Utilization Grid

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Country: Hong Kong, China





FRONT BUSBAR

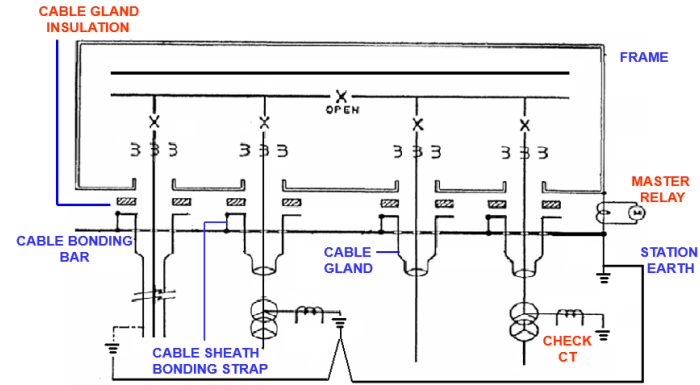
Challenge in ILOC Protection Design and Testing with Renewable Integration in High Utilization Grid

Karl M.H. LAI

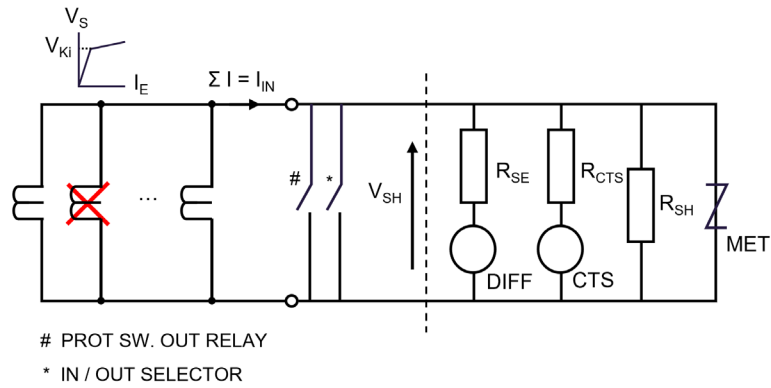
Content

- Types of Busbar Protection
- From Leakage to Frame (LTF) to Interlocking Overcurrent (ILOC)
- ILOC Protection
 - Operating Principles
 - Timeline – Busbar Fault vs Breaker Failure Protection (BFP)
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 - Tripping – DOC Guarded by OC
 - Precaution of Work on ILOC Protection and Associated BFP
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 - Evolving Fault
- Issue with Renewable Energy
 - ILOC Protection with Renewable Energy
 - Coordination with Transfer Trip (Active Anti-Islanding) and Auto-Switching (Loss of Main Issue)
- Alarm Handlinga

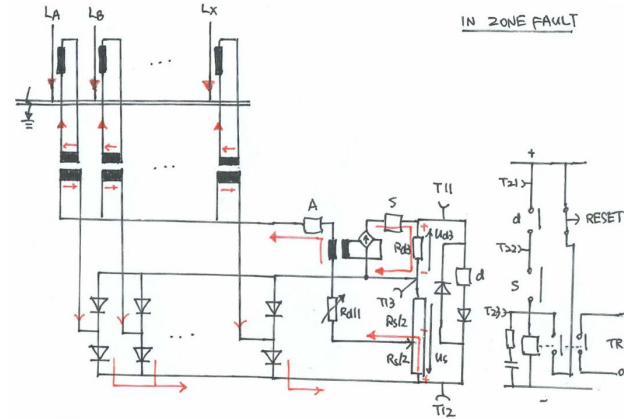
Types of Busbar Protection



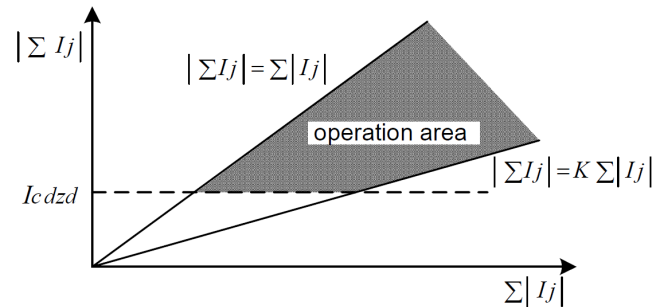
(a) Leakage to Frame (LTF)



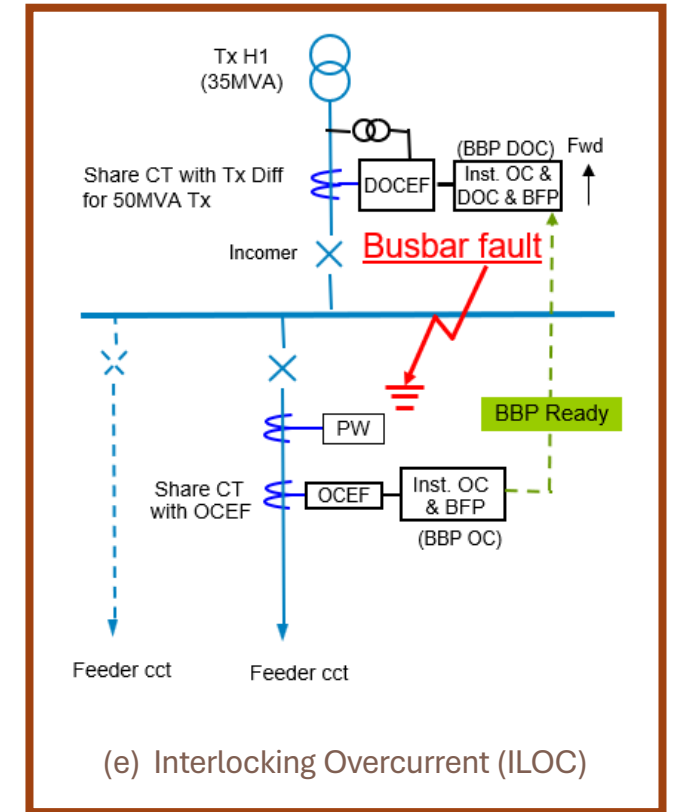
(b) High Impedance Busbar Protection (HZBBZ)



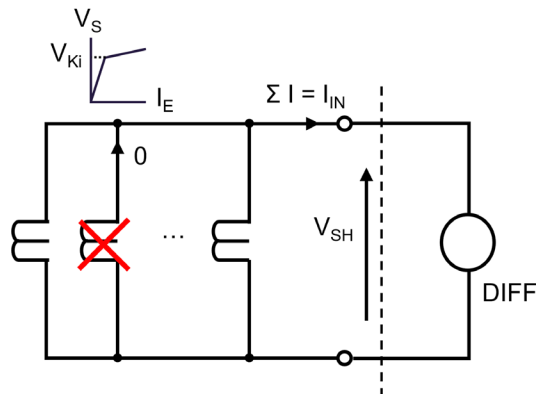
(c) Electronic Busbar Protection (EBBZ)



(d) Numerical Busbar Protection (NBBZ)

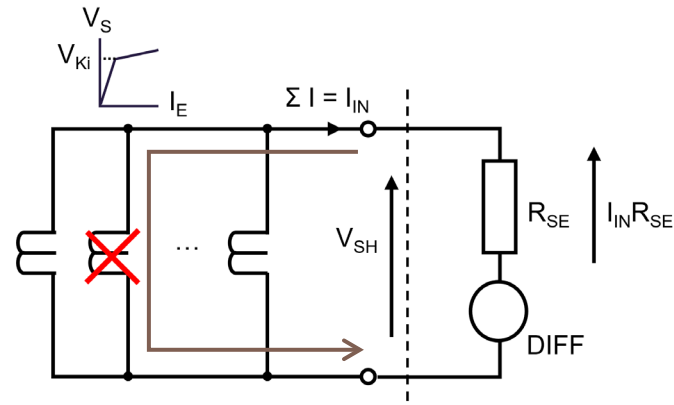


HZBBZ Scheme – Operating Principle



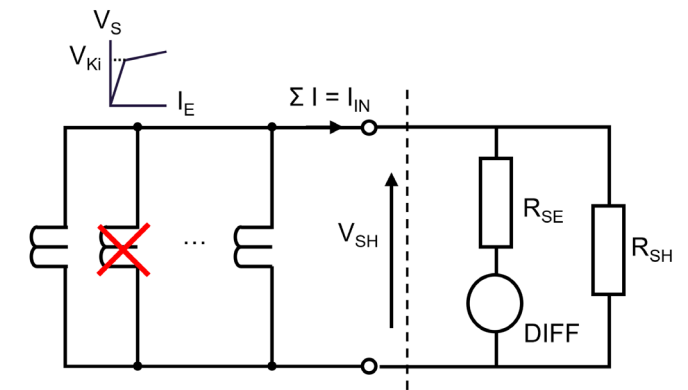
(a) CT in Parallel to Obtain Net Current into DIFF Relay

Problem – CT Saturation creates Net Current



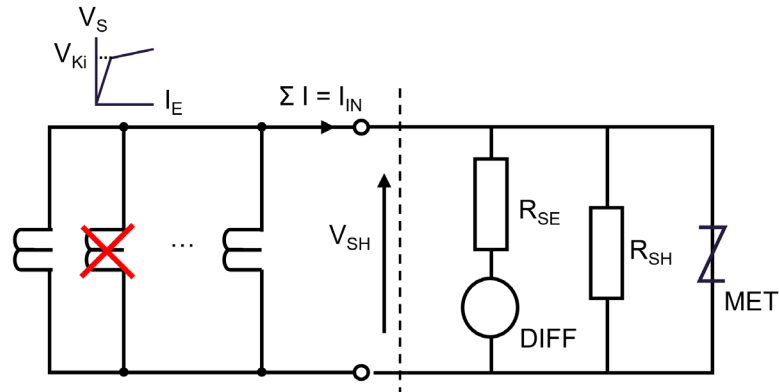
(b) High Impedance Added to DIFF path to make the differential current flow at saturated path

Problem – High Current may burn the Relay



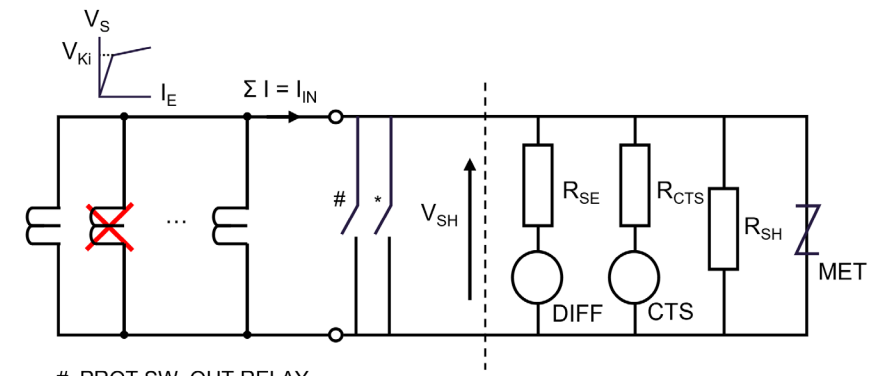
(c) A Setting Resistor Added as Shunt Path to Reduce Current into the Relay

Problem – High Voltage in Shunt may burn the Circuit



(d) Metrosil added to Short the Path at High Voltage

Problem – But the path still conducts iteratively



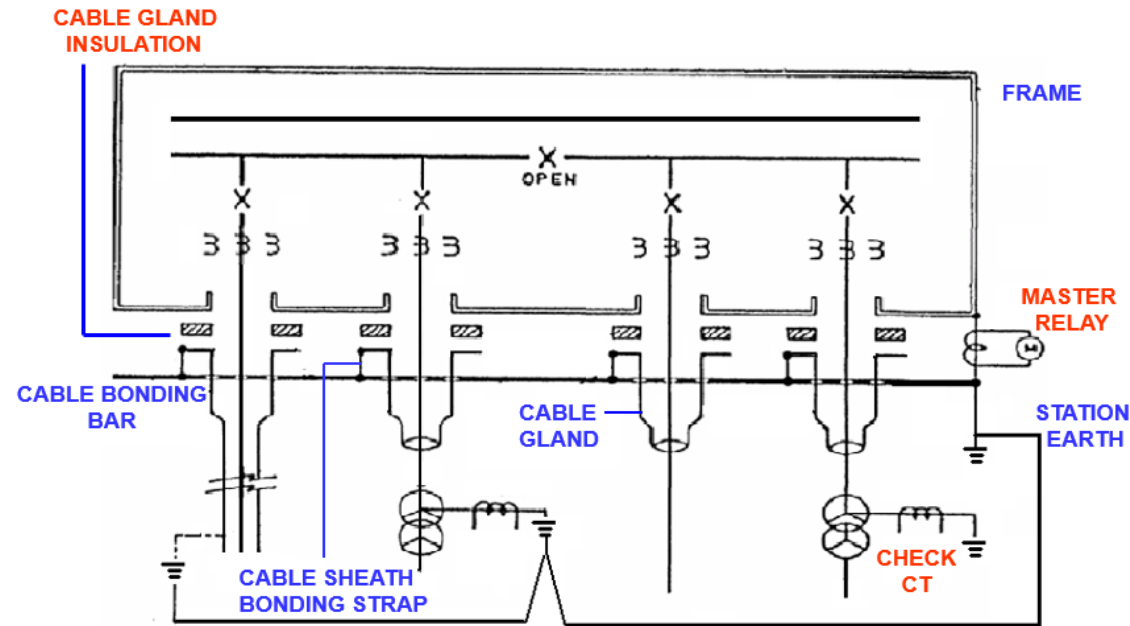
PROT SW. OUT RELAY

* IN / OUT SELECTOR

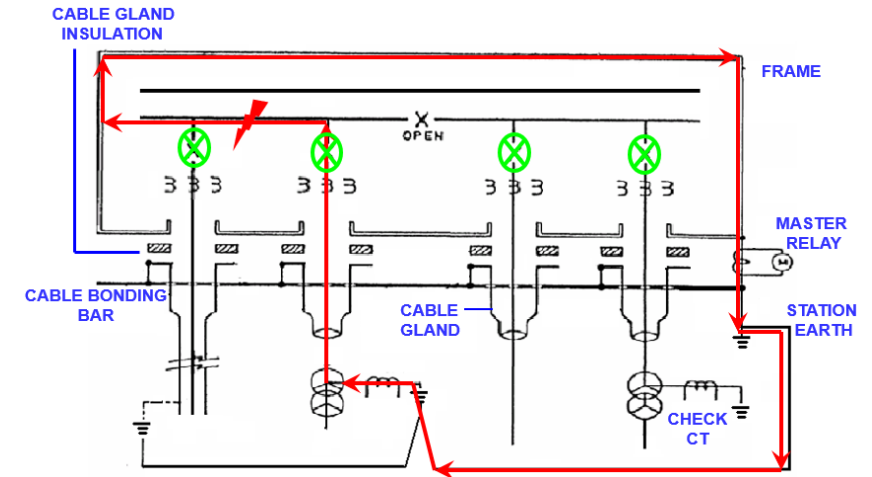
(e) CT Supervision is added to Switch OUT the Scheme by Providing a Short Path for Current Flow

Leakage to Frame (LTF) Protection

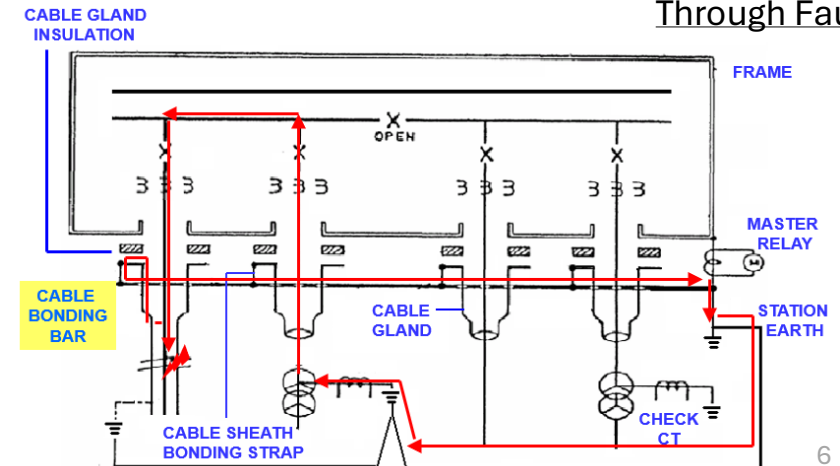
- Switchgear frame insulated from earth with 10Ω insulation material.
- Fault current between live part and frame will flow through **Master Relay** and **Check CT**, tripping all CB.
- For Through Fault, no fault current flow through master relay, LTF will not operate.



In Zone Fault



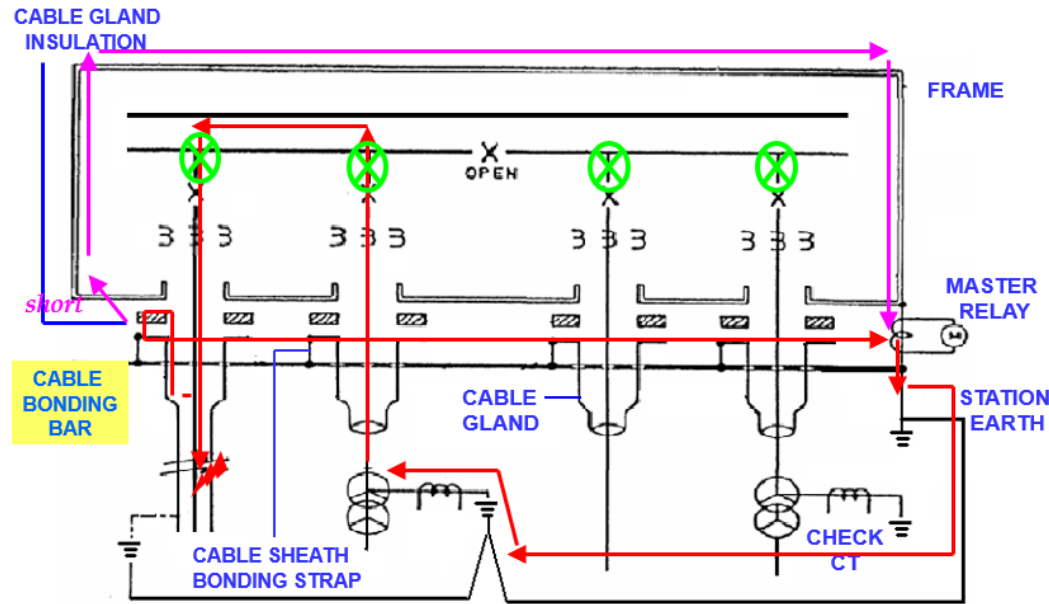
Through Fault



Unfavorable Condition with Leakage to Frame (LTF) Protection

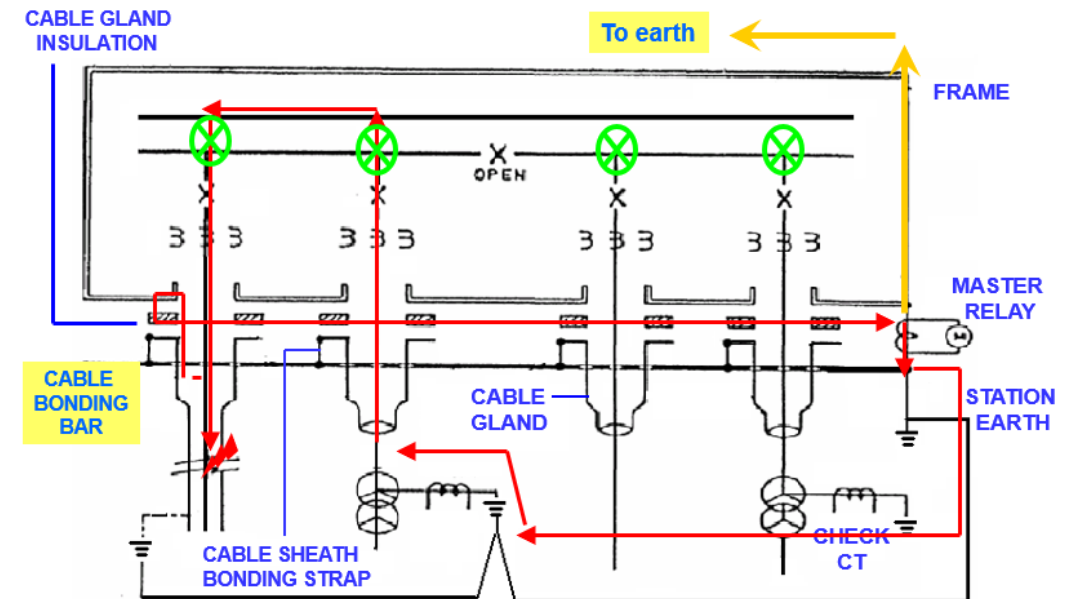
Case 1: Through Fault Unstable due to Incorrect Connection of Cable Sheath Bonding

- Originally, cable gland is insulated from frame and earthed via cable bonding bar
- Due to incorrect termination of cable sheath, or accidental earthing to frame causes LTF to mal-operate.



Case 2: Frame Earth Insulation Deterioration

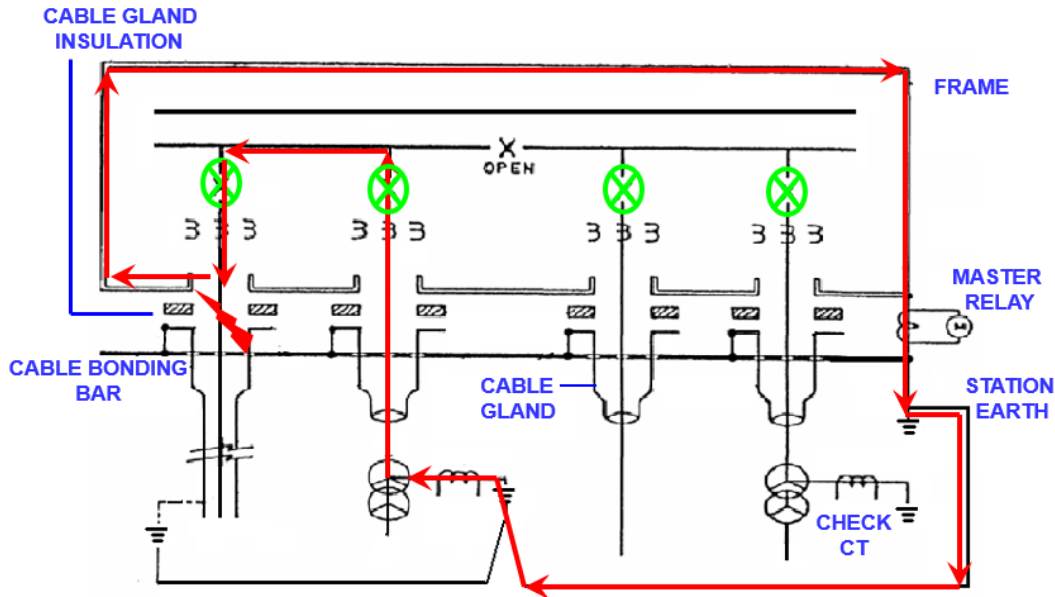
- Frame to earth insulation deteriorates due to weather and geographical conditions
- When Master Relay and Check CT sense fault current, all CB will trip
- Fault current may pass through Master Relay and earthed, although it is not designed to do so



Unfavored Condition with Leakage to Frame (LTF) Protection

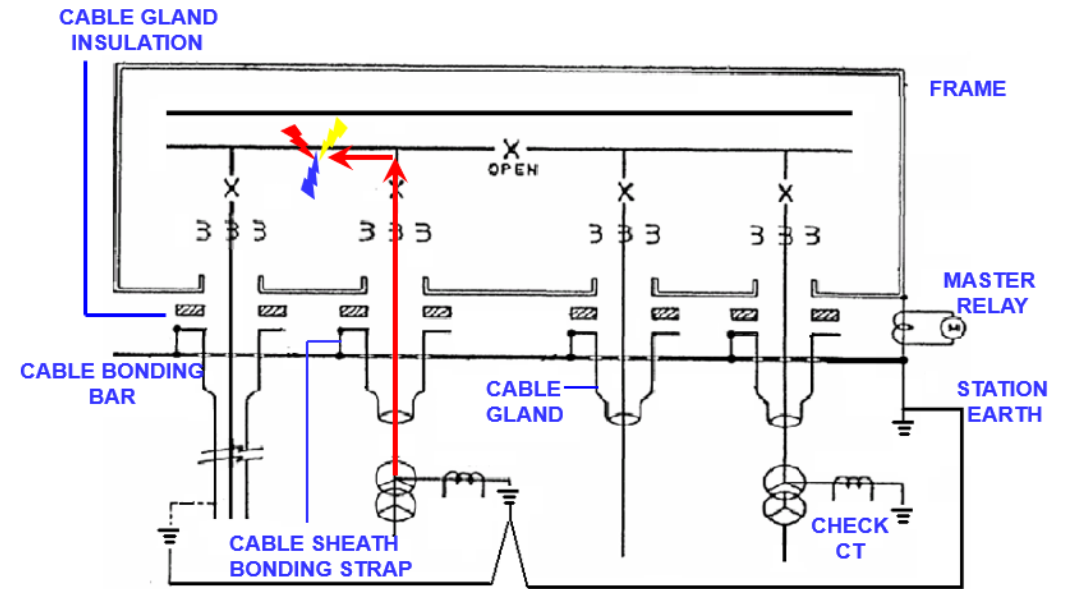
Case 3: LTF Misinterpret Cable End Box Fault as Busbar Fault

- Cable end box fault should be cleared by its own PW protection
- It possibly flashes over to frame, which makes it look like BB fault, causing LTF to operate



Case 4: Phase-to-Phase or Balanced Fault goes undetected

- Flashover to frame takes time.
- Undetected fault causes damage to system and equipment.



Interlock Overcurrent (ILOOC) Protection

- Replace LTF busbar protection
- Breaker failure protection (BFP) is integrated
- Detect all kinds of busbar fault
- Zone-discriminative (except paralleled Tx)
- Busbar fault clearance time < 300ms

Problems of LTF		Evidences
1.	Deteriorated Frame-to-Earth Insulation leading to Reduced Sensitivity	7 nos. beyond repair & thus switched out
2.	Unbalanced Earth-Fault Detection only	TEA LTF failed to operate on busbar 3-ph balanced earth-fault on 4/12/98
3.	Operative for 11kV Cable-End-Box Fault	2 incidents in past 10 years leading to LTF operation
4.	Non-zone discriminative for old LTF	HYR VCB fault on 05/01/02 led to loss of whole substation

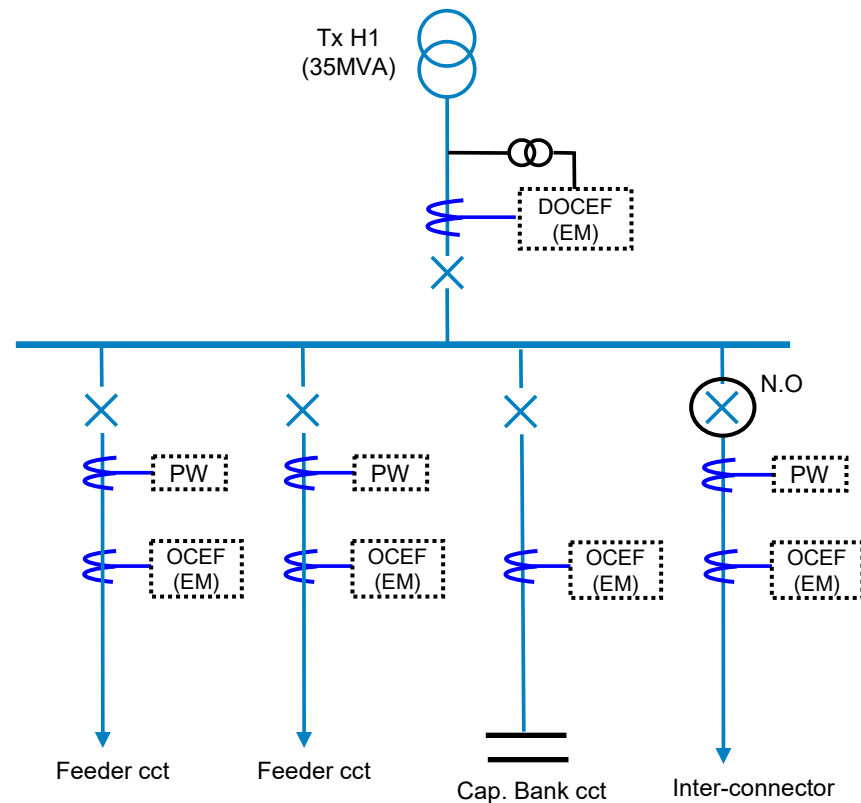
* If LTF failed to operate or switched out due to low insulation, busbar fault clearing time by HV Backup OCEF protection is very long (2.1s ~ 4.3s)

In 2003, 6 trial substations was installed with ILOC to replace LTF.

To reduce the requirement on **CT installation** and **outage**,

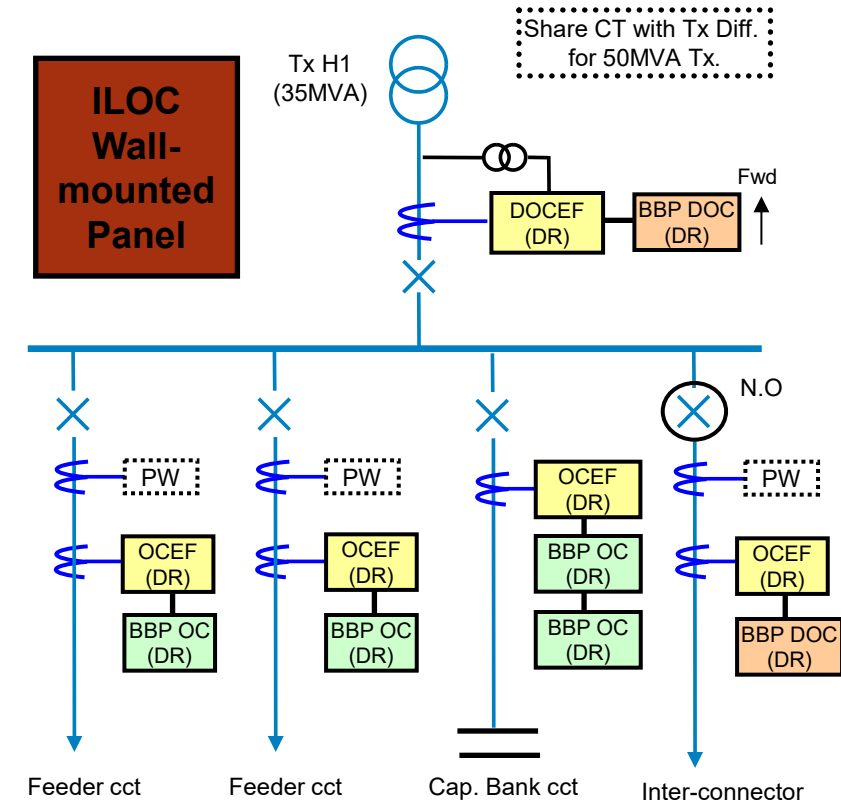
ILOC Protection is designed to make use of **directional elements** (BBP DOC at 11kV Tx and Interconnector) and outgoing feeder blocking as **interlocking element** to provide a **main protection for 11kV switchgear**.

Interlock Overcurrent (ILOC) Protection – Additional Relay Required



EM = Electro-Mechanical Relay
DR = Digital Relay

With ILOC BB Protection



BBP OC (DR) Digital Relay Models : DCD 414B
(with instant OC and BFP function)

BBP DOC (DR) Digital Relay Models : KCEG 142, DCD 914B
(with instant OC, DOC and BFP function)

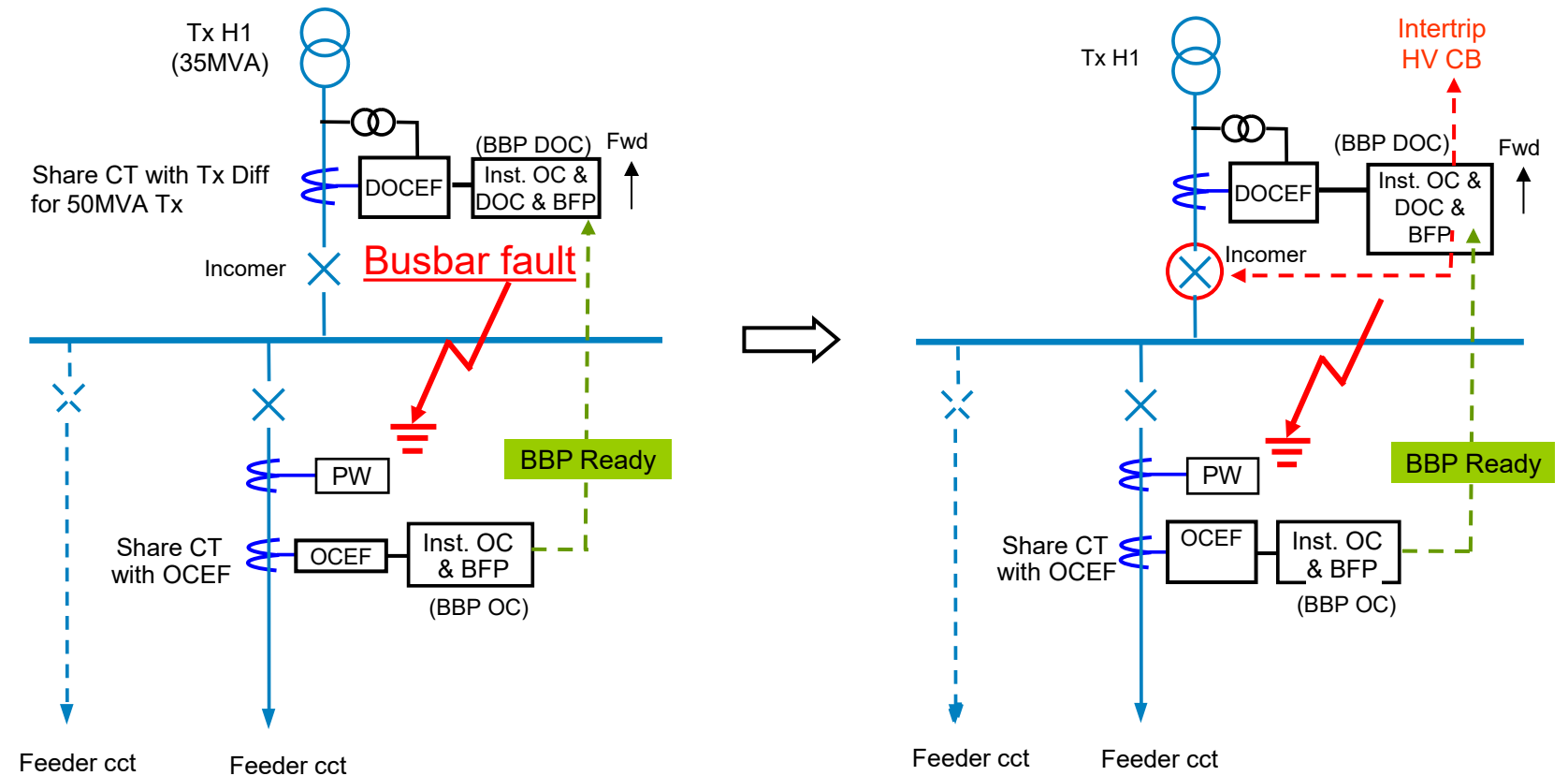
Interlock Overcurrent (ILOC) Protection – Operating Principle

Busbar Fault -

BBP DOC Relay is installed at LV side of 132/11kV Tx.

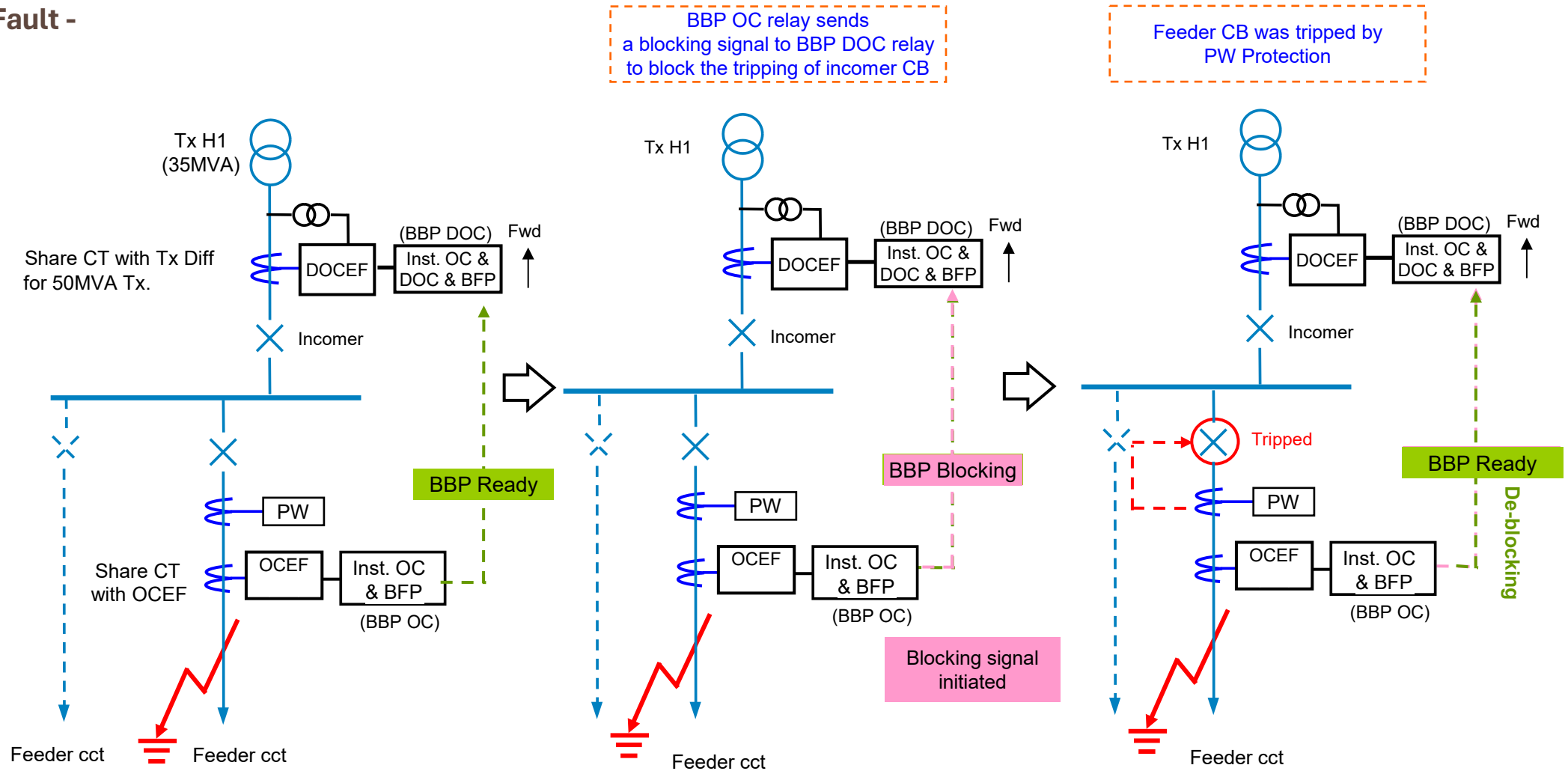
It will trip with

1. Reverse (REV, into busbar)
2. Overcurrent
(35MVA = 4500A;
50MVA = 6500A)
3. without blocking signal from outgoing feeder or other transformers.



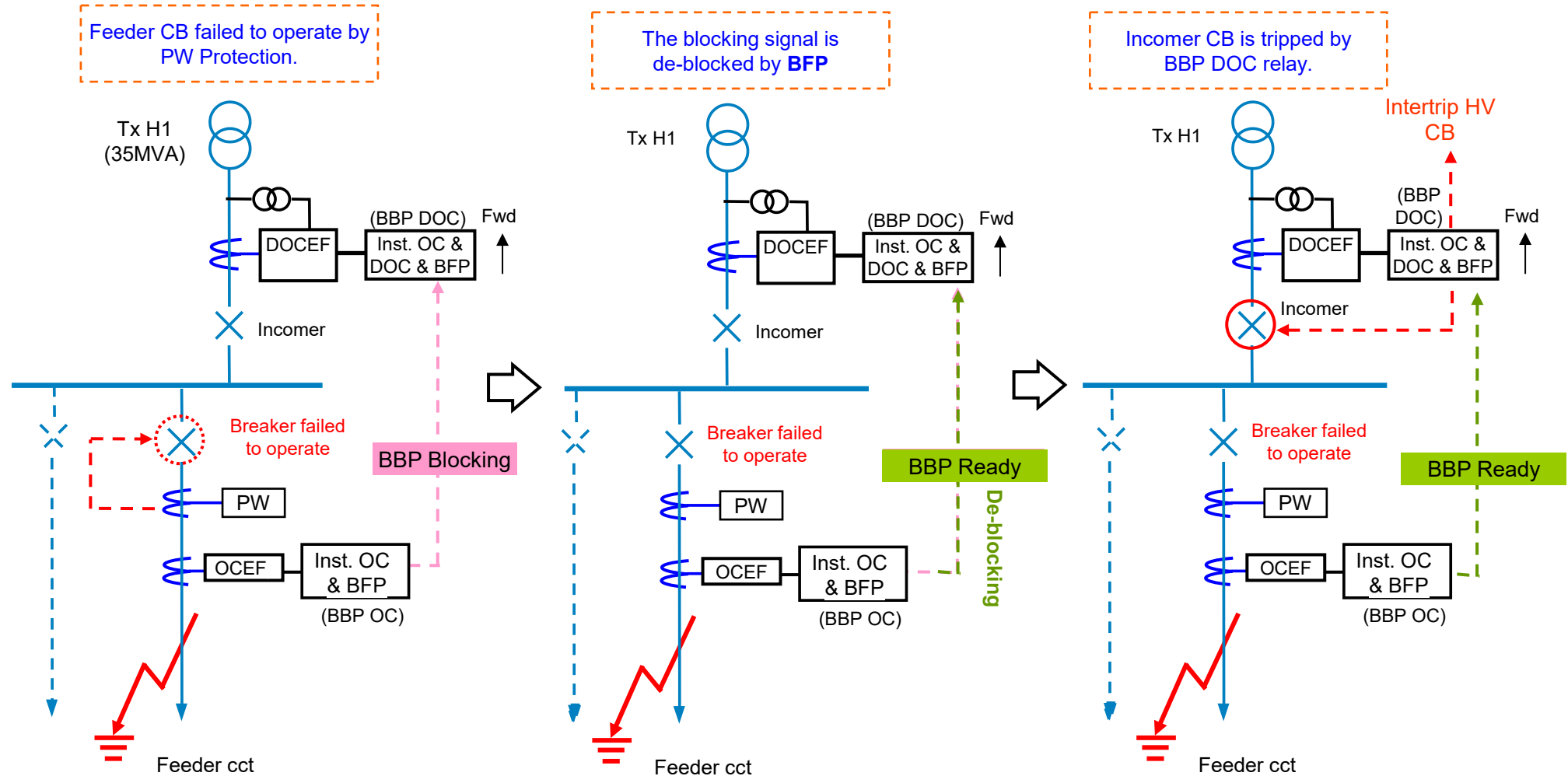
Interlock Overcurrent (ILOC) Protection – Operating Principle

Feeder Fault -



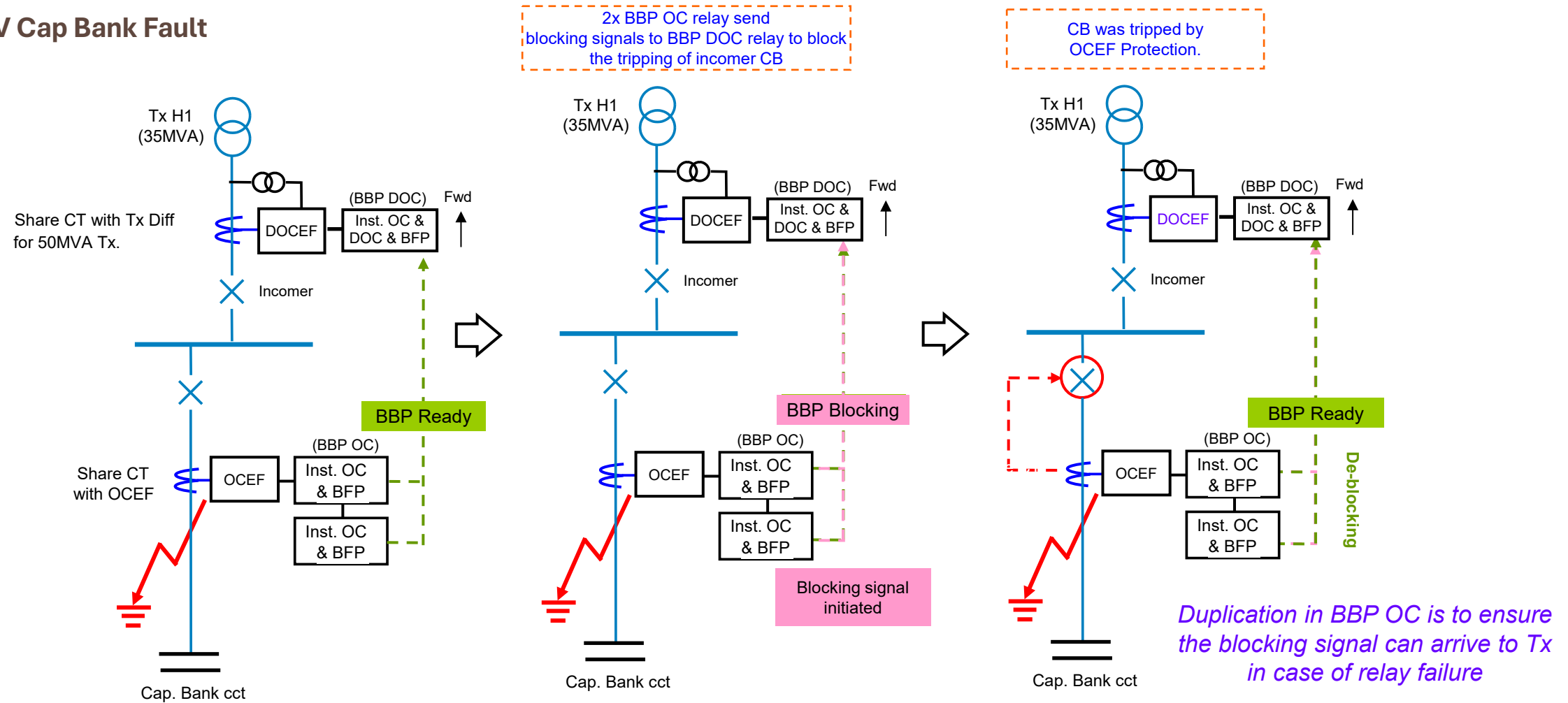
Interlock Overcurrent (ILOC) Protection – Operating Principle

Feeder Fault – (with Breaker Stuck Condition)



Interlock Overcurrent (ILOC) Protection – Operating Principle

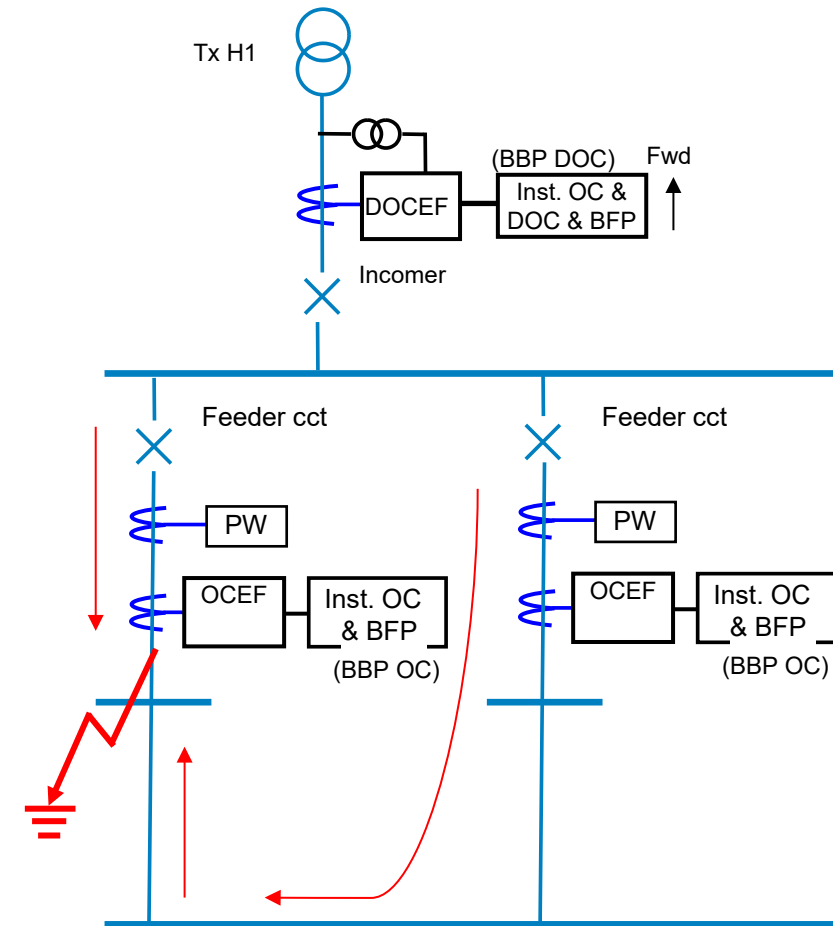
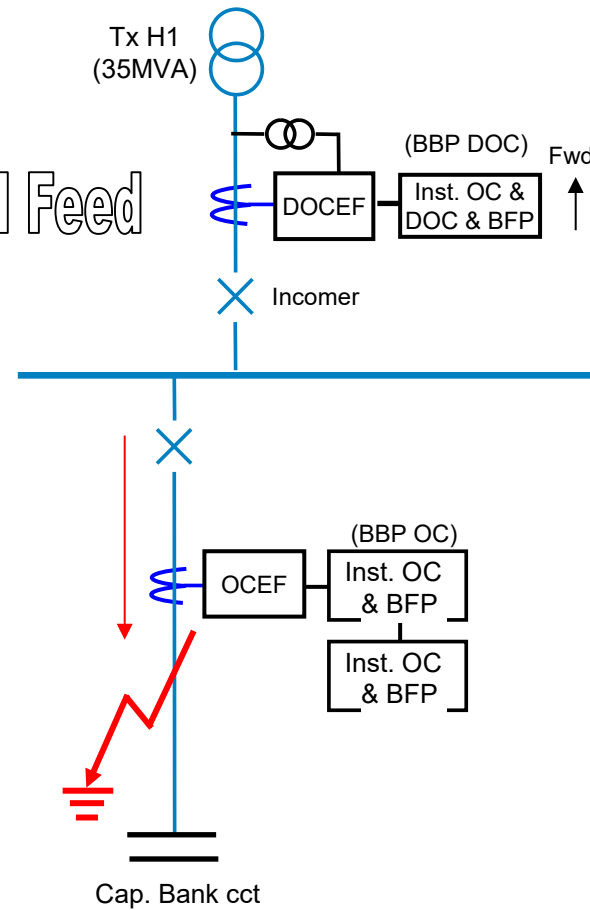
11kV Cap Bank Fault



Interlock Overcurrent (ILOC) Protection – Operating Principle

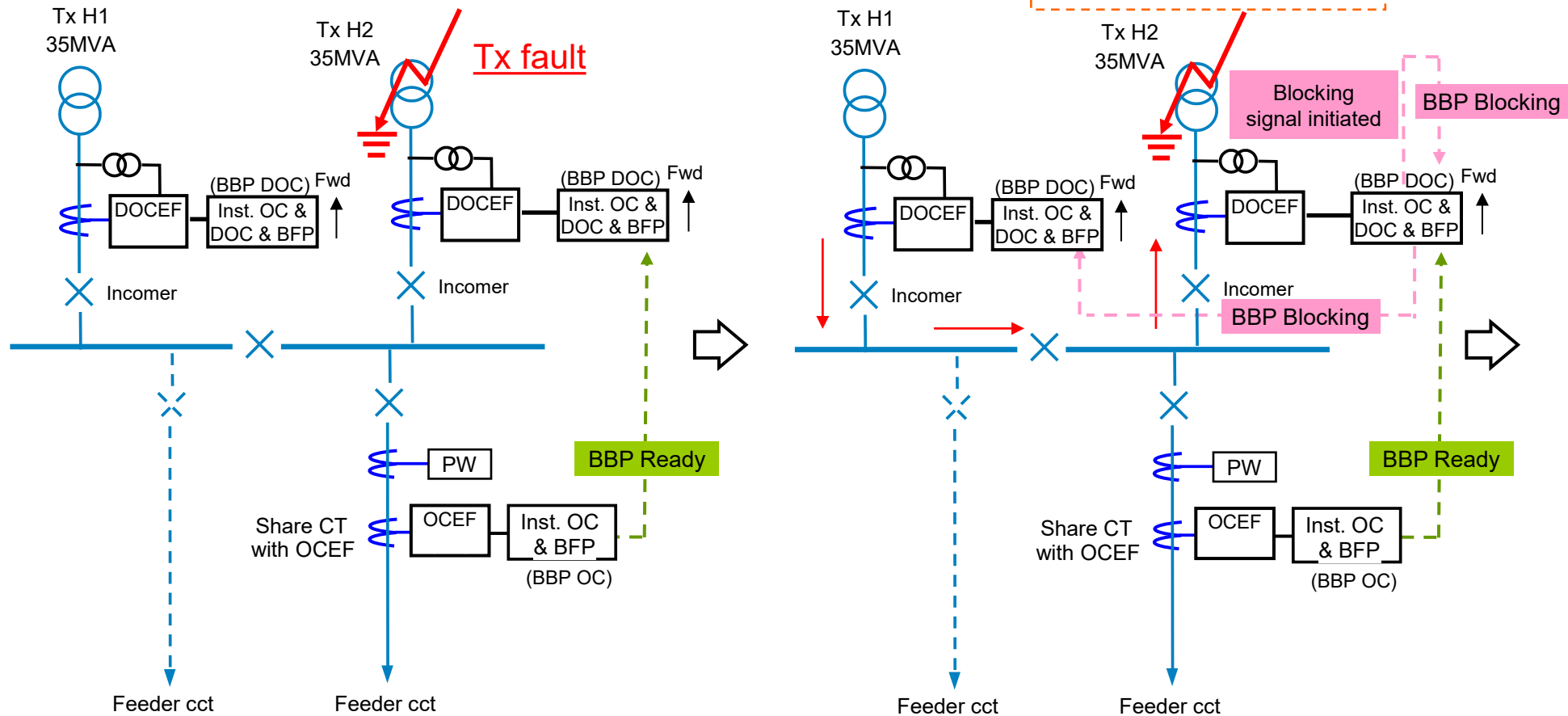
Cap Bank Fault vs Feeder Fault: Why Duplication for BBP OC at Cap Bank Circuit?

Because of Radial Feed



Interlock Overcurrent (ILOC) Protection – Operating Principle

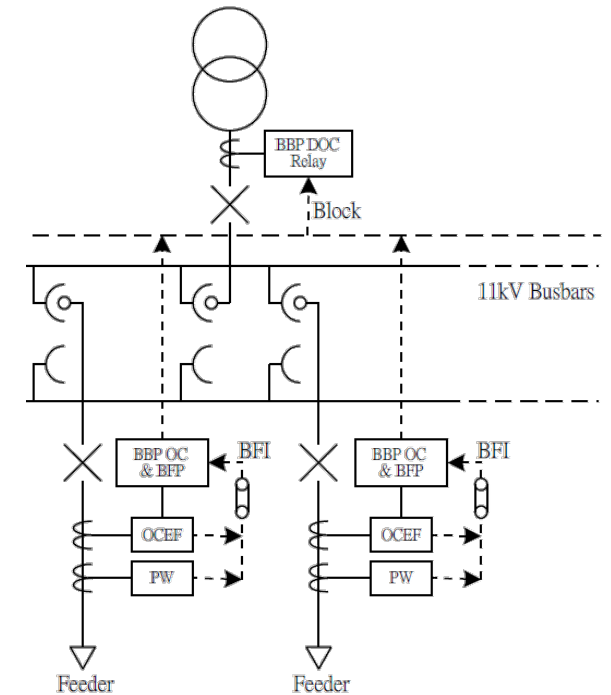
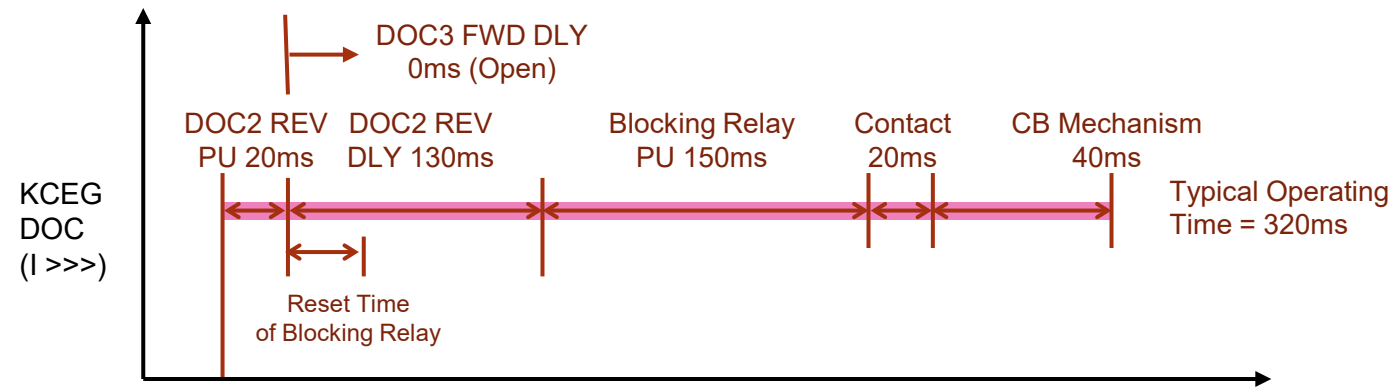
Transformer Fault under Parallel Transformer Condition



How to clear the fault if the CB of incomer under Tx fault is stuck?

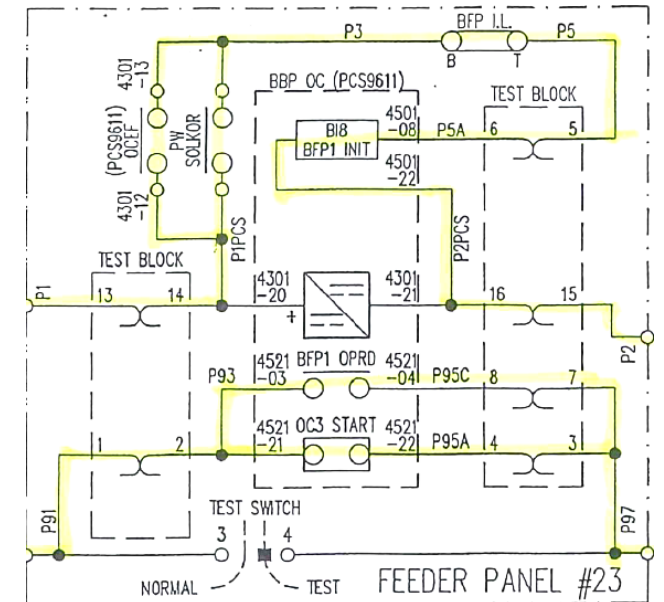
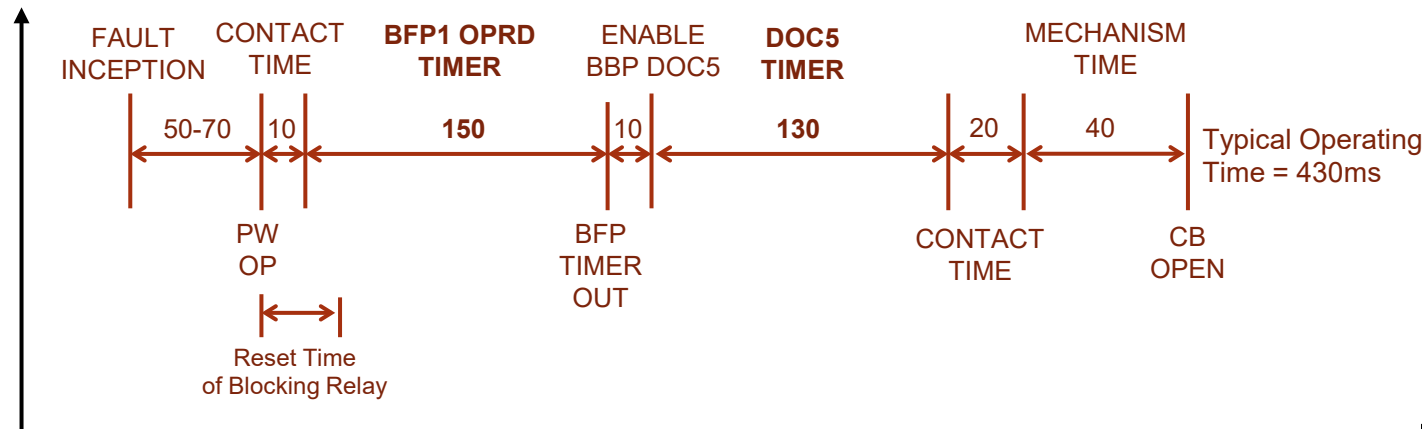
Interlock Overcurrent (ILOC) Protection – Operating Principle

Timeline for Busbar Fault Clearance:



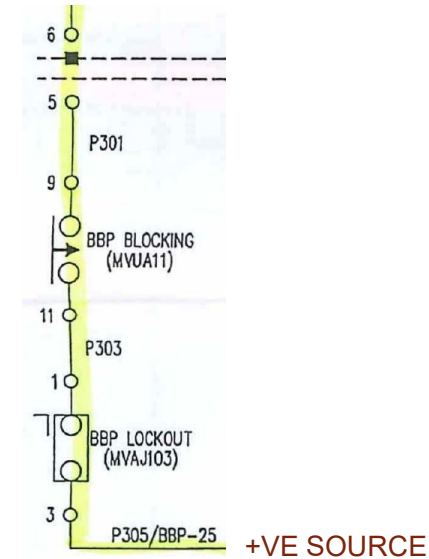
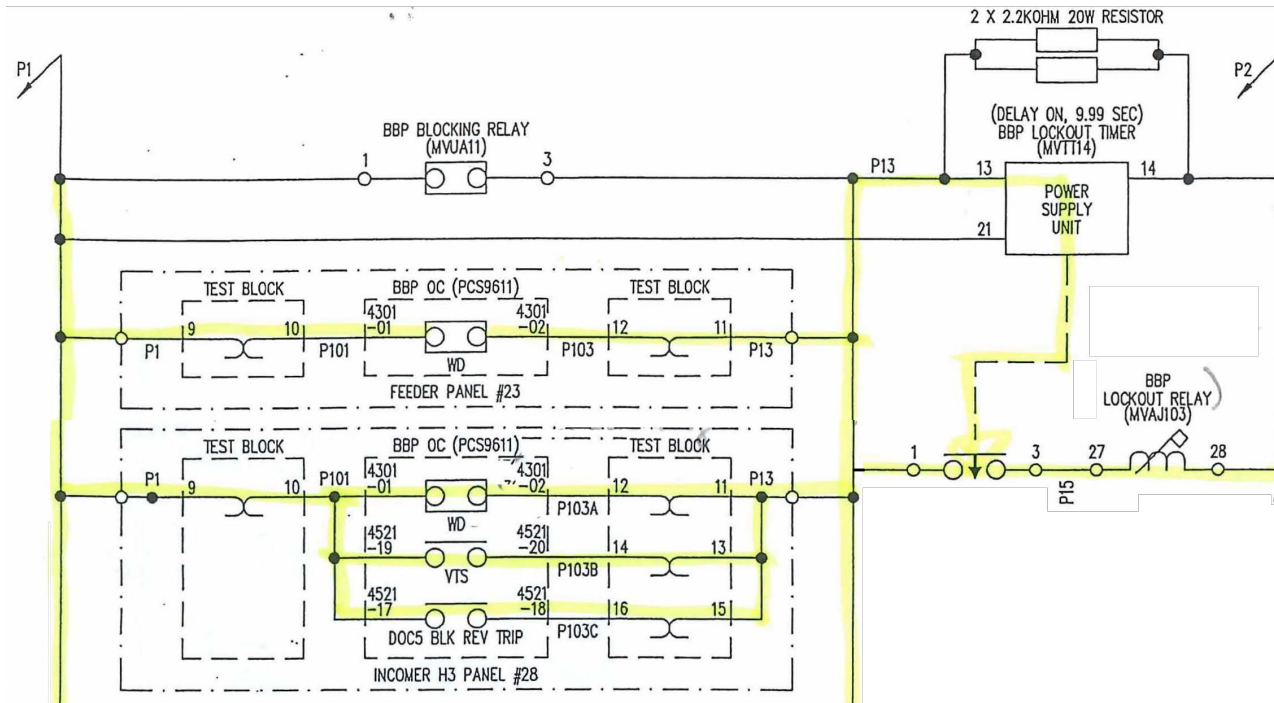
Timeline for Breaker Failure Protection for 400A Feeder Fault:

POS347: HWS011 (2016) BFP 420ms
POS362: SAH011 (2017) BFP 430ms

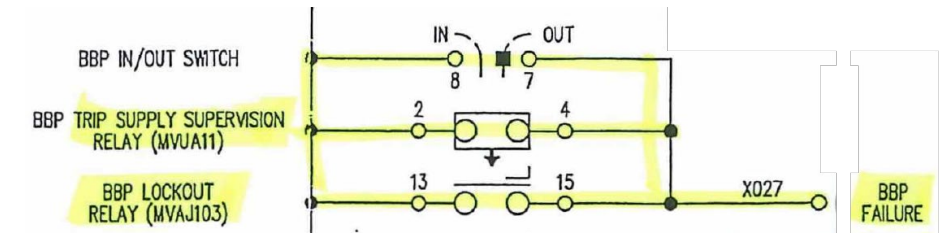


Interlock Overcurrent (ILOC) Protection – Operating Principle

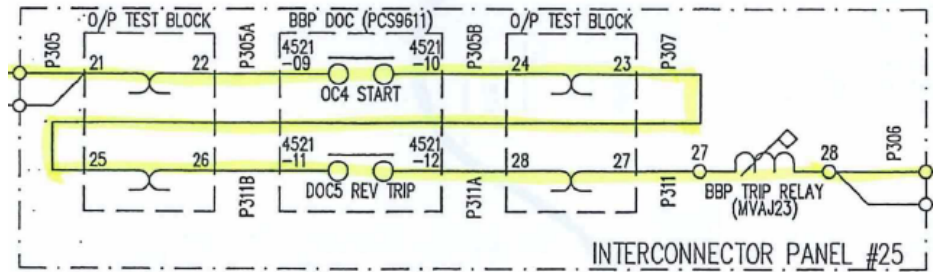
- **ILOC Lockout** defeats all ILOC tripping by disconnecting the +ve source.
- **ILOC Lockout** (Timer = 10 sec) can be caused by –
 - **BBP Blocking** (feeder OC3 START: 7.5A, 0s)
 - **WD** caused by ILOC BBP OC / BBP DOC Relay DC Failure (or CT Supervision)
 - **VTS** activated at BBP DOC (< 44V, 5s, mechanism relay dependent)
 - **DOC5 BLK REV TRIP** caused by Tx Protection Operated (inc. DOCEF)



Note – The trip supply at all incomer and interconnector panels should be continuously available. If the trip supply is lost for 5 sec, a DEF/OOC alarm can be resulted.



ILOC Protection – Element Leading to Tripping

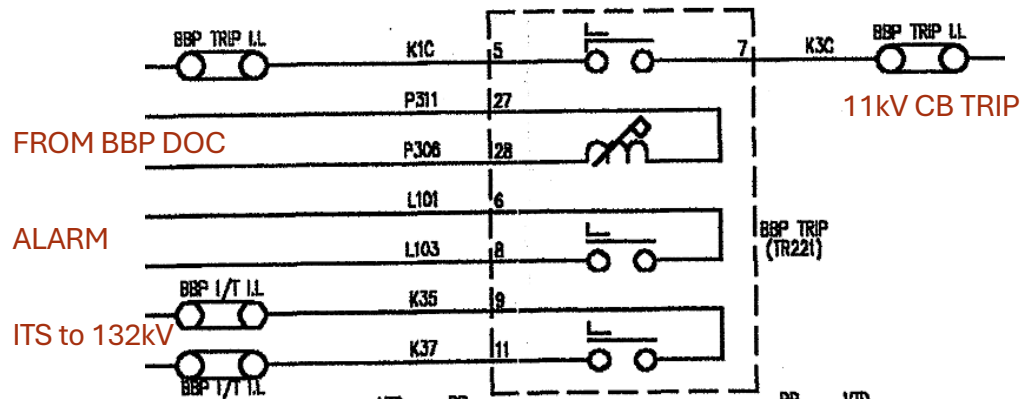


To provide a BBP trip, the BBP DOC relay at incomer / interconnector panel must have –

1. DOC2 (REV): 2400A, 0.1s contact closed to unblock
2. OC4 START: 2400A, 0s contact as OC guard closed
3. DOC5 (REV): 4500A, 0.15s contact as DOC detection closed

- (1) is to energize the blocking relay to energize the ILOC tripping path ready for tripping
- (2 + 3) is to energize the corresponding trip relay when the transformer sees the fault current.

BBP TRIP RELAY
(MVAJ23)



Further actions –

- Initiate NAS blocking (through NAS blocking link in ILOC panel)
- IF NOT RMU circuit:
Initiate ITS to trip HV side in case NOT RMU circuit,
- ELSE IF RMU circuit without CB:
Initiate RTI (CC: 0.2A, Timer: 0.3s) to trip the whole RMU ring.
- ELSE (RMU circuit with CB):
Initiate trip to HV CB + Initiate RMU CB BFP

DSOM0515 – Precautions for 11kV ILOC and Its Associated BFP

Links to remove during works on circuits in primary substations

1. **NAS Blocking Link** (Block NAS in case BBP operate, remove blocking link to allow restoration in case BBP operated due to work)
 - Secondary Injection Test of OCEF (Interconnector)
 - Change of BBP OC Relay Setting
 - Modification and Re-commissioning of ILOC BBP Scheme, e.g. busbar extension, energization of cable circuit before ILOC on load test.
2. **BFP Isolation Link** (provide BFP signal to ILOC scheme to allow tripping of Tx LV CB under feeder fault; remove to avoid BFP initiation, but ILOC lockout after 10sec injection)
 - Secondary Injection Test of CD or OCEF (Feeder) – to avoid tripping the incomer CB in case of a cable fault
 - Secondary Injection Test of OCEF (Interconnector)
 - Stability Test for PW / CD

Section 4.2 - OCEF (Feeder) Injection:

- **Keep current injection time < 10s**
- Select “BBP Test Switch” to TEST Position with consent from SCE (NOT recommended when circuit on load)

Section 4.3 - OCEF (Interconnector) Injection:

- **Remove Trip Isolation Link (+ BFP & NAS Blocking)** – BBP DOC may operate depending on phase relationship of current injection and VT signal.

Section 5 - Stability Test for PW/CD (Feeder):

- **Defeat PW Protection** (e.g. remove PW trip isolation link) to maintain load current for the test
- **Remove BFP isolation link** to avoid tripping the incomer CB in case of a cable fault (BBP DOC relay operate faster than feeder OCEF)

Section 9.3 – Busbar Extension:

- **Issue TLR** to prohibit ILOC BBP scheme when ILOC scheme is modified.

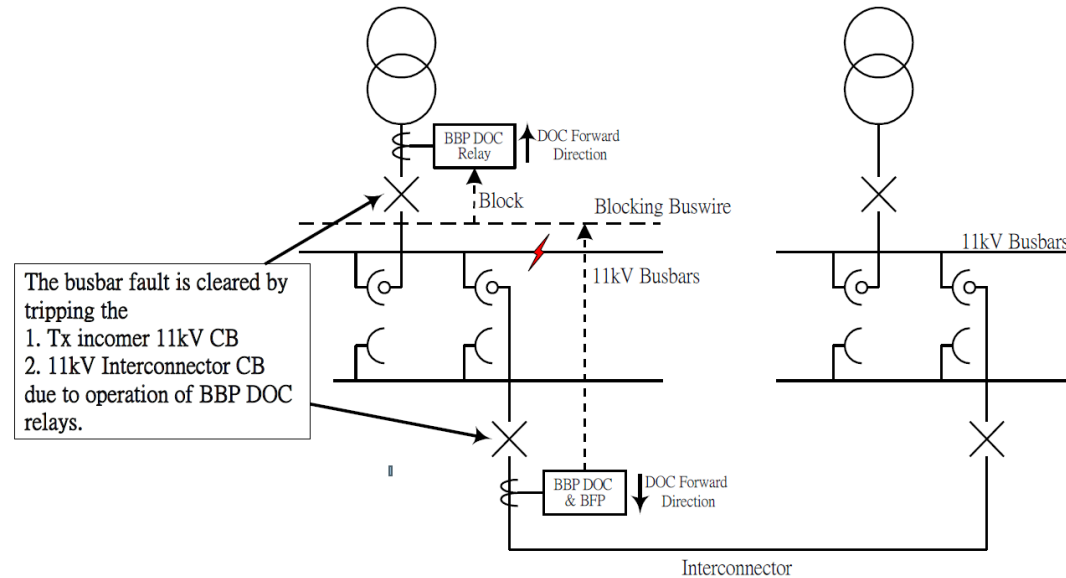
Section 9.4 – Terminating Cable to Spare Panel

- ILOC scheme remains *switched in* during terminating cable
- Before energization, **remove BBP trip and intertrip isolation links** of transformer incomer and interconnectors **and NAS Blocking Link** on ILOC BBP wall mounted panels.

DSOM0515 – Precautions for 11kV ILOC and Its Associated BFP

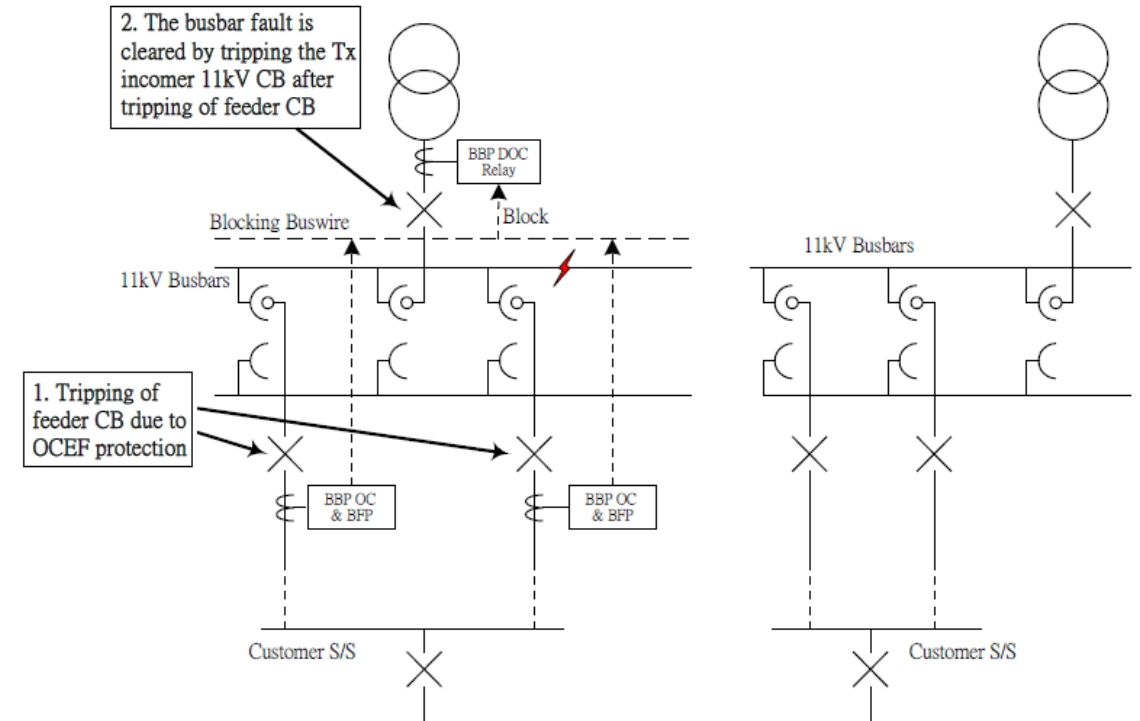
Section 7 – Parallel Operation with Other Primary 11kV Interconnectors

7.1 Parallel Transformer through Bar-to-bar Interconnector



Busbar Fault will be cleared by BBP DOC of Tx and Interconnector (as BBP DOC is faster than OCEF in feeder)

7.2 Parallel Transformer through Ring Interconnector



1. Feeder BBP OC sees the fault current → Block BBP DOC
2. Feeder OCEF trips feeder CB.
3. BBP DOC of Tx unblocked → Trip Tx LV CB through ILOC

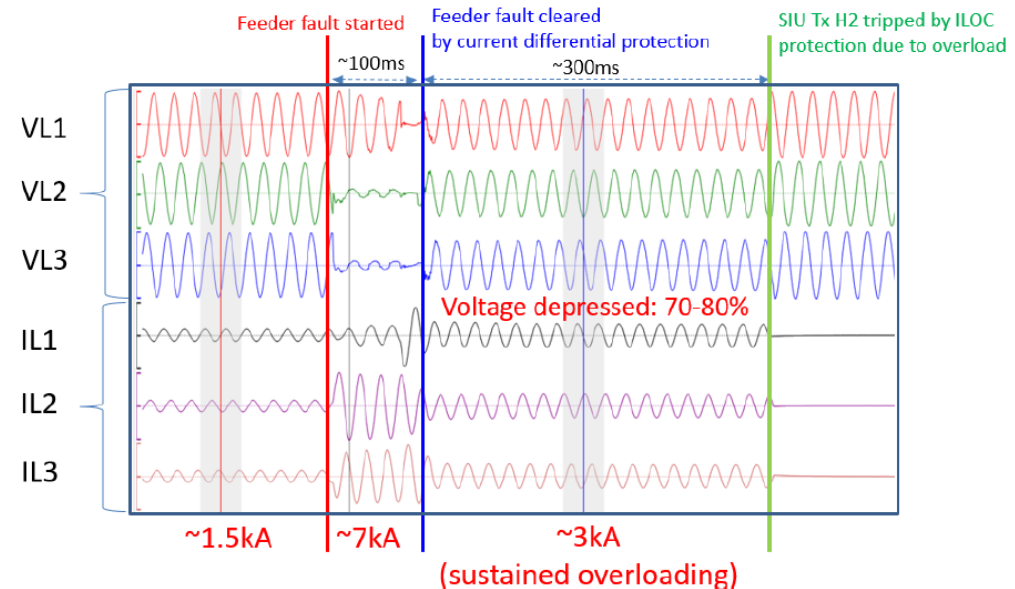
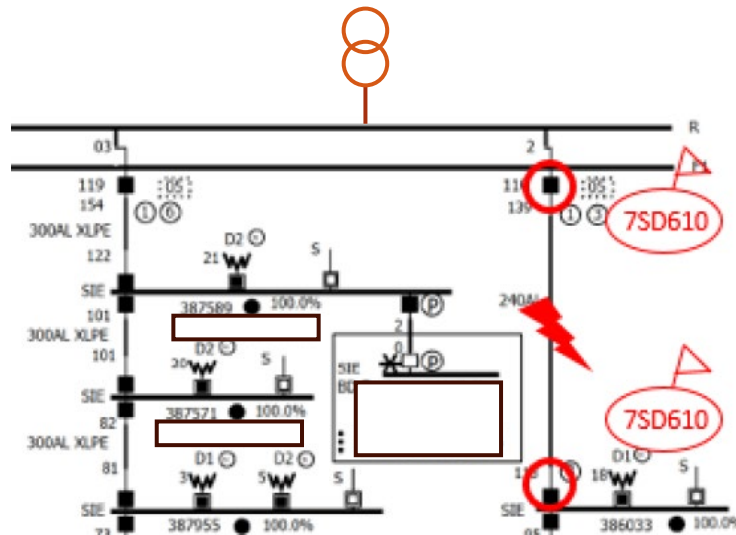
POS373 – ILOC tripped with Stalling Current

SIU011 – Phase 1 High Set Tx OC Setting

- A **genuine cable fault** occurred on a 11kV circuit in SIUR0005. The respective CD relay (typed 7SD610) operated to clear the cable fault with 0.1sec.
- Immediately after the fault clearance, there was an **abnormal rise in load current** due to **stalling current** in the SIU supply area, leading to overloading of the 132/11kV transformer (Tx) to 160% of its rated value, with supply voltage dropping to 70-80% of the rated voltage. ILOC busbar protection operated and tripped the CB of SIU Tx H2 leading to supply loss of Rear Busbar in SIU011.
- The overload was suspected to be caused by a large number of air-conditioners restarting with **excessive stalled current** after a severe voltage dip. The outgoing feeders were **only slightly overloaded, insufficient to block ILOC busbar operation**, which then protected the transformer from exceeding its cyclic rating.

To prevent recurrence, immediate actions include:

- De-sensitizing ILOC busbar protection to provide more margin for Tx overload current after voltage dips.

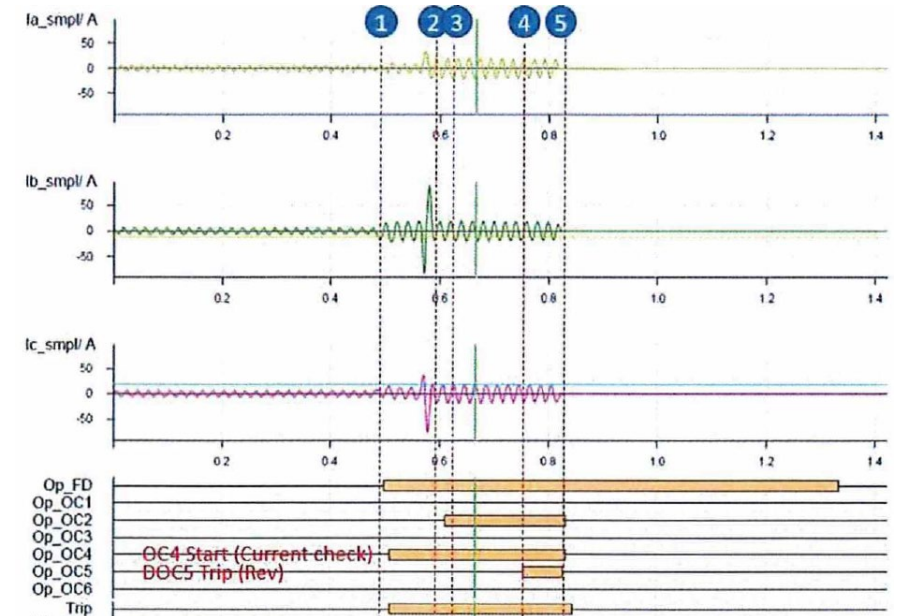
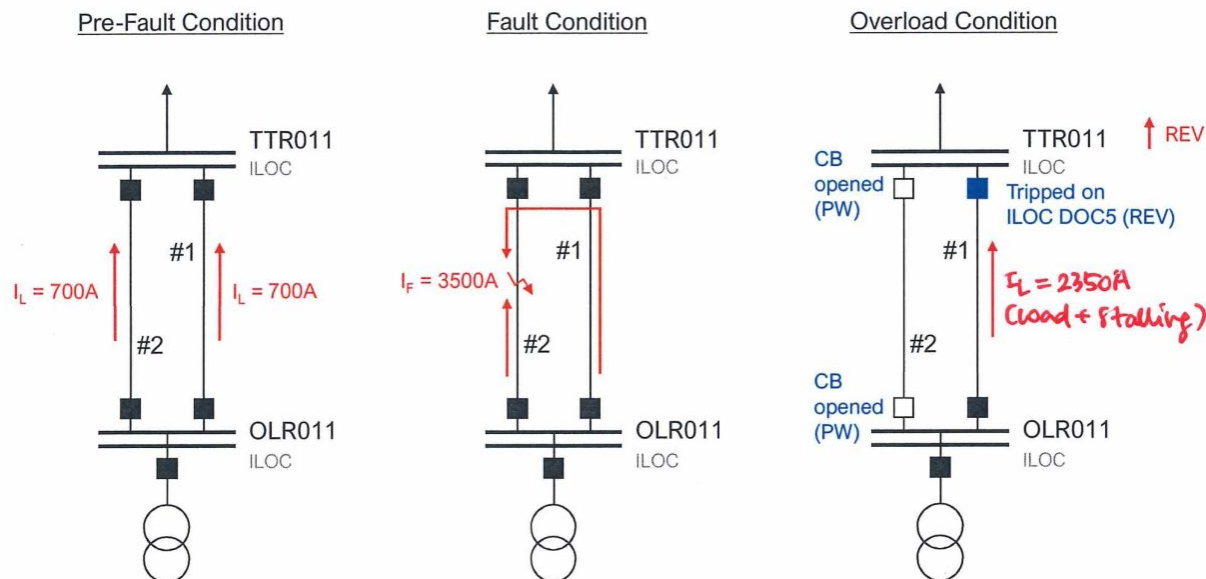


POS386 – ILOC tripped with Overload

OLR011 – Phase 2 High Set Interconnector OC Setting

ILOC Issue: Can be tripped due to overload or high stalling current.

- The loadings of interconnector OLR – TTR No.1 and No.2 (full rating 800A) were both around 700A prior to the incident.
- Investigation revealed that interconnector No.2 was on cable fault, and the circuit correctly tripped on PW differential protection in around 100ms to clear the cable fault.
- After the fault clearance, all loadings were shifted to interconnector No. 1. **Load current** and **post-fault stalling current** were raised up to around **2350A**, which could be regarded as overload current to interconnector No. 1. The original design of ILOC is mainly for protection against busbar fault, while it can also provide a bit of overload protection in certain condition.
- As the cable fault at interconnector No.2 was cleared, the Forward Directional Overcurrent element (**DOC3 FWD Start**) of ILOC BBP DOC relay on Panel 22 no longer operate to block the ILOC scheme. Therefore, with **DOC5 REV** setting of 1520A (190%) & 0.13s, the “Current check element” (**OC4 Start**) and “Reverse Directional Overcurrent Trip” (**DOC5 REV Trip**) then operated after around 130ms, and the BBP Trip relay picked up and operated to trip Panel 24 CB as engineering designed.

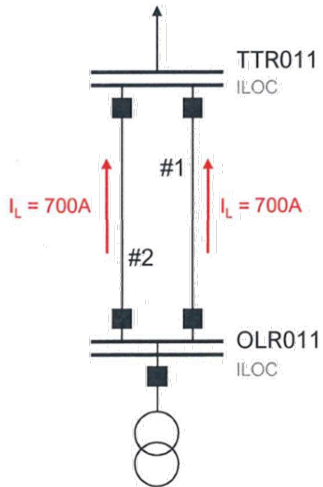


SCI 230008: ILOC DOC Setting for 11kV Bar-to-Bar Interconnector

After the incident of OLR – TTR No.1 & No.2 **ILOC Overload Trip** (POS386), SCI 230008 was issued to change the setting of DOC under **unfirm condition** -

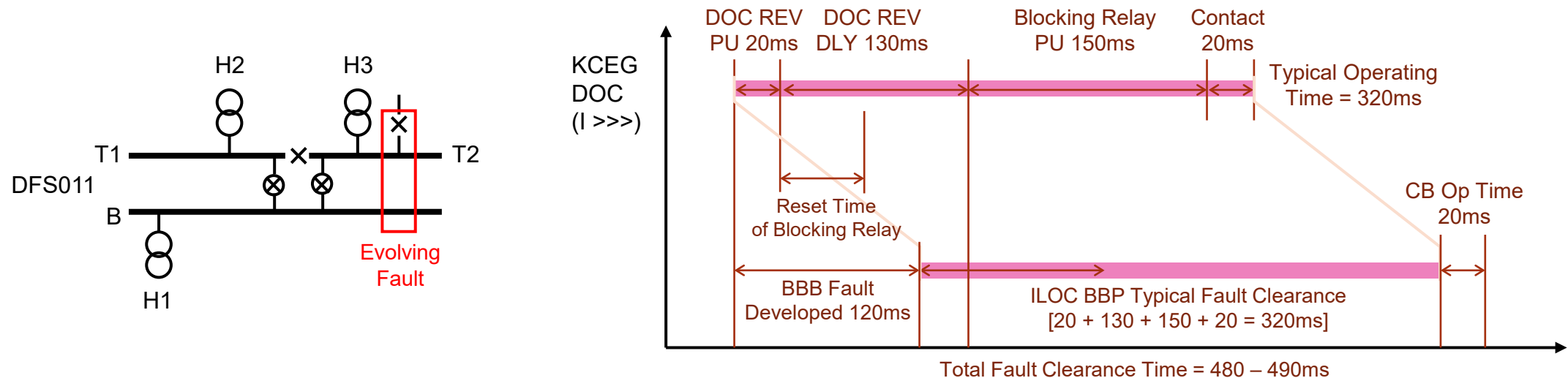
- Existing DOC settings were for **firm conditions** (2 x 800A interconnectors under 800A total).
 - Under unfirm condition –
 - Adjust **ILOC BBP DOC relay settings at the load side** to prevent ILOC BBZ operation during overload.
 - Allow the remaining overloaded circuit to trip by **circuit OCEF protections** at both ends.
 - Increase default ILOC BBP DOC setting was not preferred due to reduced busbar protection sensitivity. Recommend **temporary change** of ILOC BBP DOC setting on a need basis.
 - Pre-calculated temporary settings for future situations, generally **2.5 times the predicted total overloading current** when **one interconnector faults**.
 - To minimize the chance of high impedance BB fault unable to trigger ILOC BBP, consider **running the 35MVA transformers in parallel** to **maximize fault current**.
 - Restore the setting as soon as the risk of overloading is mitigated.
- Example:** When **1400A** is flowing from OLR011 to TTR011 through 2 x 800A interconnectors and TTR011 has ILOC BBP, the temporary IC DOC setting for the two interconnectors at TTR011 (load end) would be **2.5 x 1400A = 3500A**. No setting change at OLR011 side (source end) is required.

Pre-Fault Condition



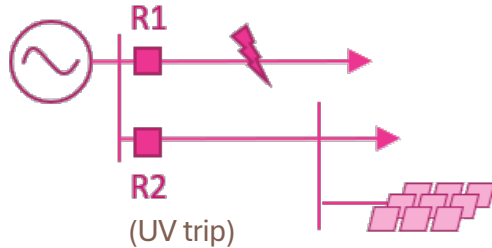
POS375 – ILOC Delayed Blocking with Evolving Fault

- Site investigation revealed that Panel 25 CB connected to T2BB was on fault. Since T2BB was supplied by Tx H2 and H3 running in parallel.
- “Overcurrent (OC) element” ($I >>$) of ILOC BBP DOC relay of Tx H2 and H3 detected the busbar fault current and picked up at around 20ms to block the ILOC scheme first.
- Normally, for such a BB fault scenario, “Rev DOC element” ($I >>>$) of the same relays would operate and de-block the ILOC scheme 130ms later (i.e. 150ms after fault occurrence), eventually clearing the fault at around 360 - 370ms
- However, in this case, at 120ms after T2BB fault occurrence, the fault further developed to BBB which was supplied by Tx. The Tx H1 BBP DOC relay “OC element” ($I >>$) picked up in 20ms (i.e. 140ms after T2BB fault occurrence) and kept blocking the ILOC scheme for 130ms more. Therefore, while the BBB fault was cleared in 360ms as expected normally, the T2BB fault was cleared in 490ms (120ms + 370ms) under this multi-fault scenario.



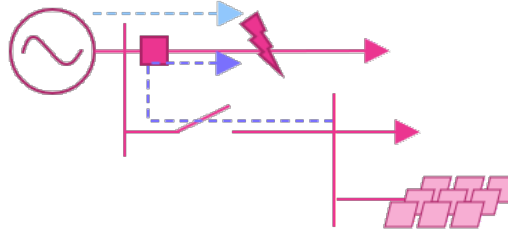
How does Renewable Energy affect current protection practice?

1. Sympathetic Tripping due to DG type, location & capacity



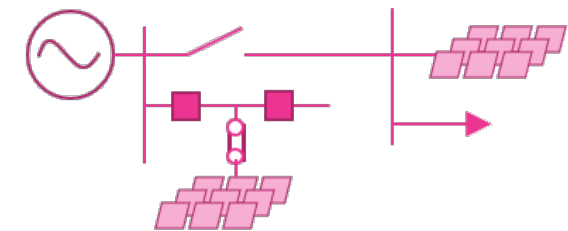
- Dynamic OC Setting
- Directionality for Blocking
- Transfer Trip

2. Blinding in Protection with DG



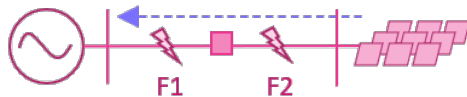
- Adaptive OC Setting with/ without DG
- Check CT Location
- Reduce source impedance dependent protection

3. Auto Reclose Coordination in Protection with DG



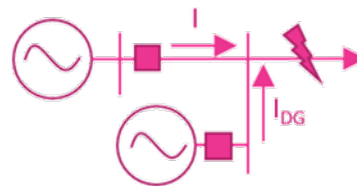
- Replace fuse / recloser with adaptive relay
- Delay reconnection time (= 5 min)
- Avoid leading asynchronous operation of machines

4. Bidirectional Fault Current due to DG (> Load)



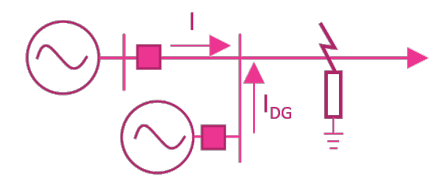
- NOT necessary from downstream to upstream, can be **Random flow** in graphs
- **Directional** with interlock
- Current Differential/ Permissive Overreach (Unit Protection)
- **Nodal & Zonal Protection**

5. Reach Setting in Distance Element



- $Z_{seen} = \frac{V}{I} = R_L + \left(1 + \frac{I_{DG}}{I}\right) R_F$
- **Overreaching Effect** and Delayed Tripping
- **Admittance + IDMT** or **Directionality** to identify internal/external fault

6. High Impedance Fault due to insulation failure

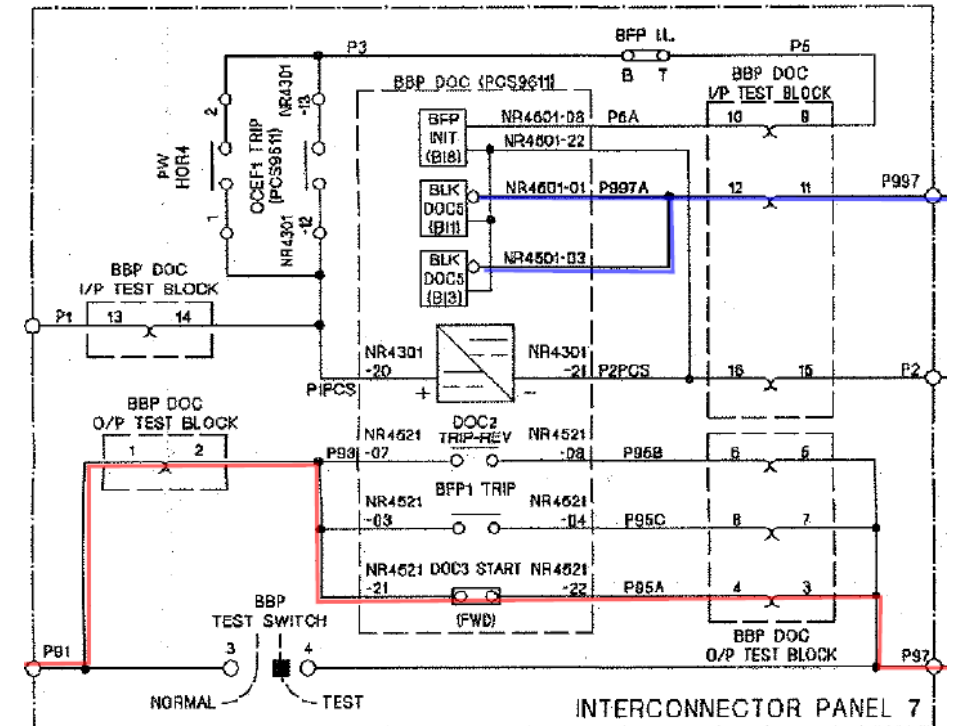
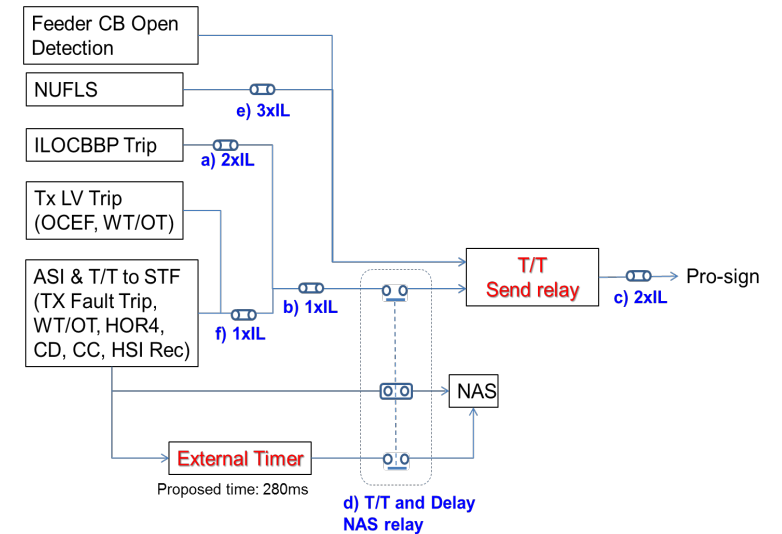


- Possible causing **thermal fault**
- Arc Fault by **arc fault detector**
- **Asymmetric** in current waveform due to breakdown voltage
- Current magnitude escalates gradually until reaching constant

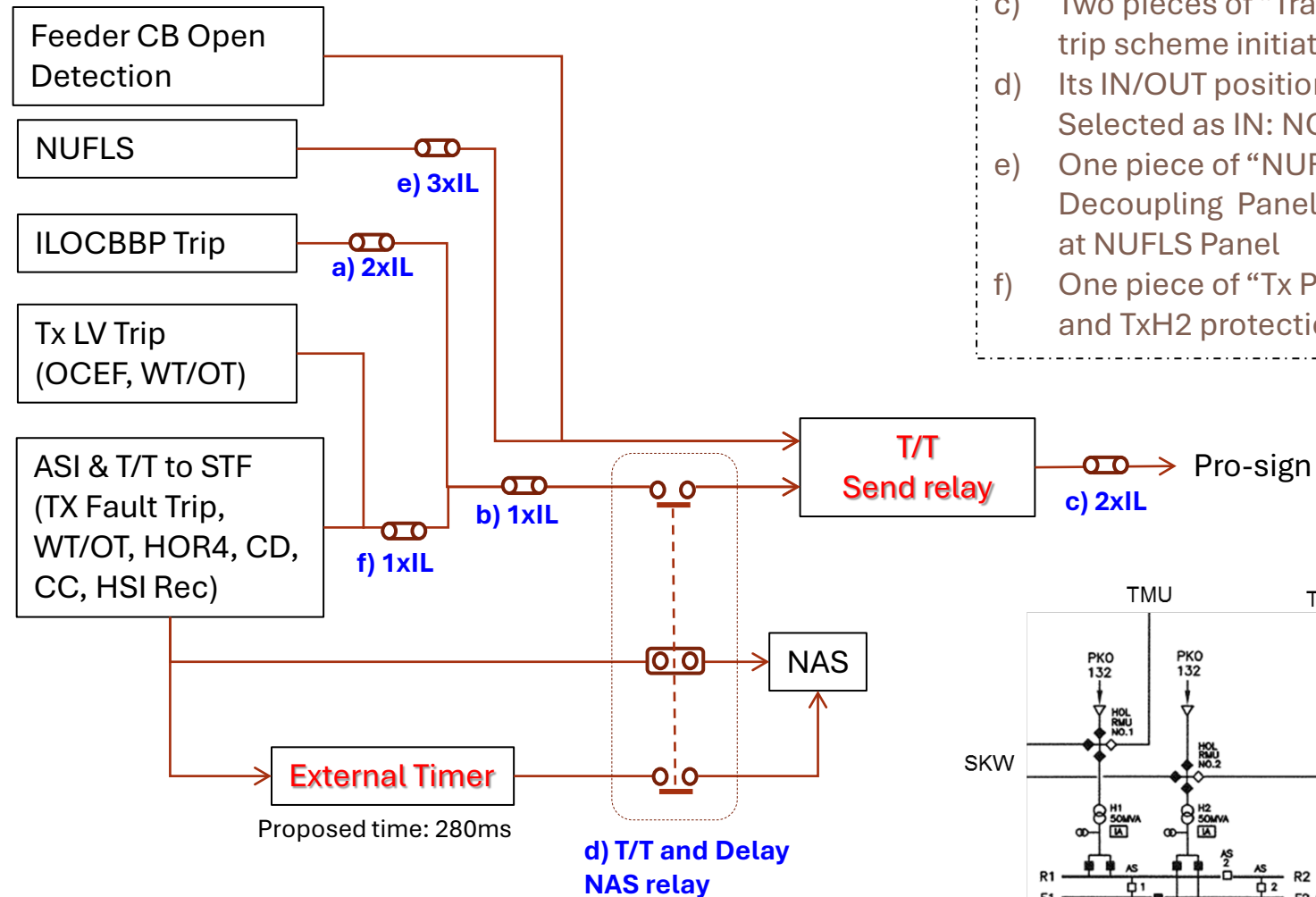
ILOC Protection with Renewables (e.g. HOL – STF)

To provide interconnectors for renewables, the ILOC scheme should be amended –

- From **feeder** configuration to **interconnector** configuration –
 - Include **Voltage Selection Scheme** with MVAP22 to transfer voltage from VT @ Transformer panel for polarizing
 - Include **Transfer Trip Function** to ILOC (Section 3.4.13)
 - Include **Re-trip Function** to Local CB for tripping (Section 3.4.12)
 - Include **BBP Tripping function** with DOC5 [REV] and additional trip relay (Section 3.4.2)
 - Include **BFP Tripping function** from PW / CD and OCEF relay at Local Panel (Section 3.4.8)
 - Include **blocking** from other circuits (P997, Section 3.4.4)
 - Include other elements such as **DOC3 START** as forward blocking to provide block signal to others (Section 3.4.1)
 - Include **VTs function** to lockout ILOC (Section 3.4.10)
 - Include **DOC5 REV BLK function** to lockout ILOC (Section 3.4.11)
- Carry out Function Check on **NAS Blocking**

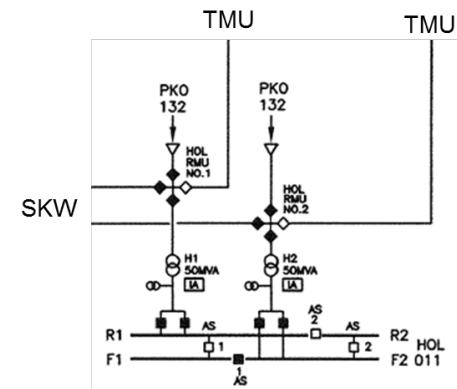


Transfer Trip (TT) and Delayed NAS

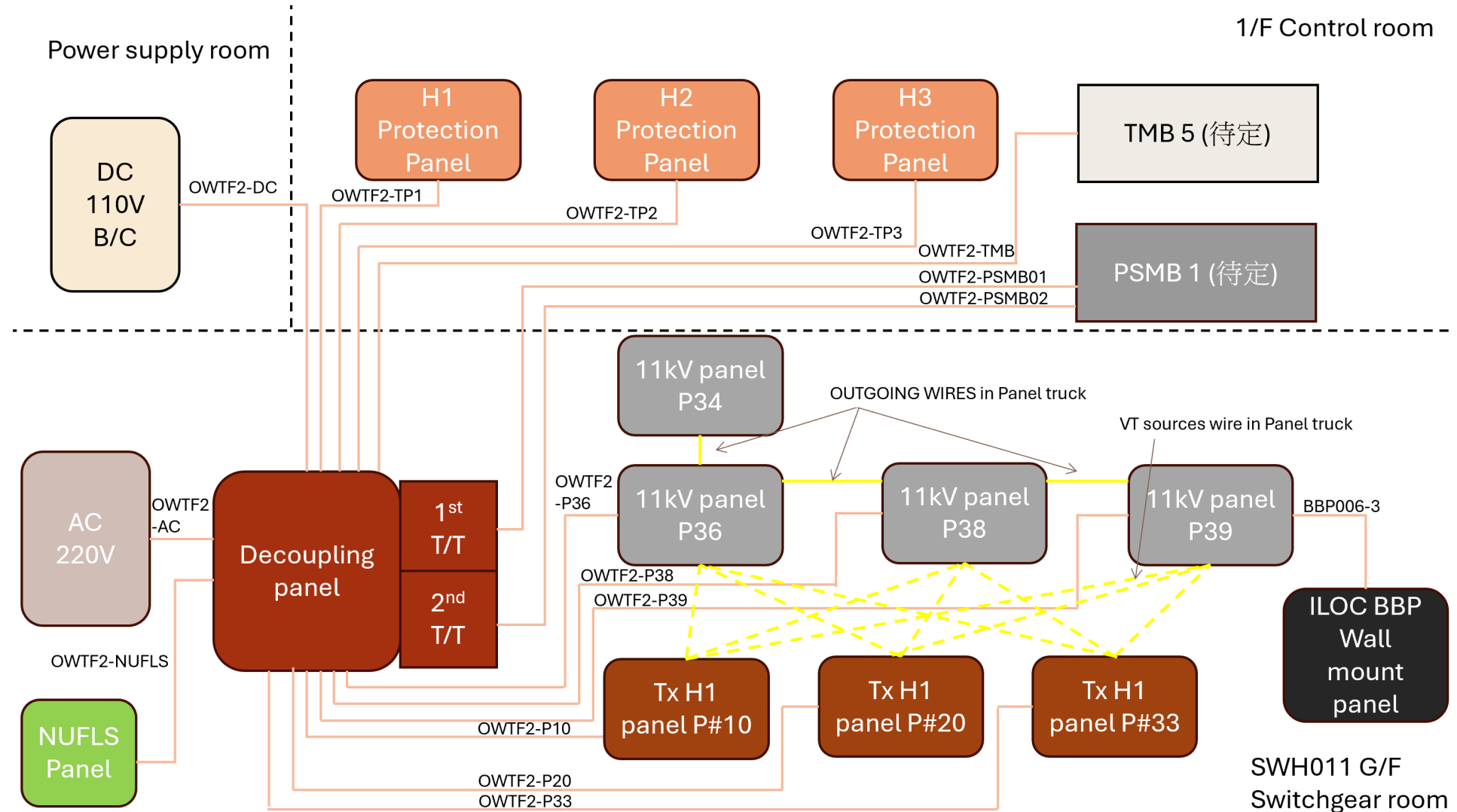


Remarks

- Two pieces of “T/T to STF Isolation Link” at switchgear panel TxH1 and TxH2 for Transfer Trip to STF initiated by busbar protection
- One piece of “Tx T/T to STF Isolation Link” for both Tx H1 and Tx H2 at STF System Decoupling Panel
- Two pieces of “Transfer Trip to STF Isolation Link” for transfer trip scheme initiated at STF System Decoupling Panel
- Its IN/OUT positions can be selected locally or remotely. Selected as IN: NO contact closes while NC contact opens
- One piece of “NUFLS T/T to STF Isolation Link” at STF System Decoupling Panel and two pieces of “T/T to STF Isolation Link” at NUFLS Panel
- One piece of “Tx Protection T/T to STF Isolation Link” at TxH1 and TxH2 protection panels



Additional Wiring Required From Feeder to Interconnector (RE)



BB (ILOC) DEF/OOC Alarm Handling

Issues	Reason
Loss of VT Voltage [UV1 Operated LED lit on]	• VT Secondary Fuse Burnt • Interconnector: MVAP22 Voltage Selection Scheme not successfully transfer voltage to BBP DOC relay → Often activated upon switching • VT Input for BBP DOC relay fail Note 1: Measure test block voltage to check equivalent to relay measurement to determine if VT fail or relay VT input fail Note 2: Perform Protection Isolation with the consent of SCE.
BBP OC / DOC Relay Failure	• No Healthy LED or With Alarm LED Activated → Hand Reboot • Relay Failure with Relay Shutdown [TNL Incident from Karl] → Relay Replacement or Power Supply Module Replacement • Relay Stuck at Startup Page or with WD activated → Relay Reboot • CT or VT Module Failure [OLR Incident reported from Liu] → Check misalignment between input and relay measurement → Relay Replacement
Continuous Injection	• OC activated to disconnect P91 – P97 under injection lead to blocking
Defective Supervision Relay	• MVUA cannot be hold for drop-off 5 sec [HOL incident reported from Karl]



THE END

ANY QUESTIONS?

Power System Analysis and Protection Studies

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